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Exploring the extent to which player modelling and personalisation impact the capacity of video games for the development of emotional intelligence

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Abstract

Video games offer unique affordances for engagement not seen in other media. This report summarises existing literature in the spaces of player modelling, emotion theory, personalisation and emotional intelligence to provide a conceptual link between these four fields and assesses the effect that personalisation has on the potential of video games as a tool for prolonged emotional regulation. This report designs and conducts a study which evaluates the impact of player modelling and personalisation inclusion when measuring differences between self-reported trait emotional intelligence (EI) scores. A video game prototype was developed and provided for participants to play - this prototype asked participants several questions and used the answers to present a personalised sequence of challenges. Two sets of data were gathered through a pairing of surveys and an optional semi-structured interview. When processed and statistically tested, numerical data did not show compelling results to suggest a strong link between player modelling/personalisation and increased trait EI scores. It is strongly suggested that this stems from drawbacks in the design of the game rather than the underlying theory which stands as the basis for this project. When looking at qualitative interview responses, several points of insight were highlighted for future iterations of this study and research into affective games.

Statement of Originality

I, Rostislavs Popovs, declare that this report, ’Exploring the extent to which player modelling and personalisation impact the capacity of video games for the development of emotional intelligence’ and the work presented in it are my own. I confirm that:

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- I did all the work myself and have not directly helped anyone else. Where guidance was provided in the form of group supervisor meetings, academic integrity has not been breached.
- The material in the report is genuine, and I have included all my data and designs.
- I have not submitted any part of this work for another assessment.

Signed: Rostislavs Popovs

Date: May 9, 2023

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  Their work is provided under a Creative Commons Zero license.
  Their asset pack 'Ultimate House Interior Pack' was used within this project.

Contents

1.1 Motivation

Throughout the past decade research on interactive media has shifted from the potential harm of video game engagement towards the positive emotional and developmental effects of engaging with the medium[1][2]. This shift highlights video games as a means of emotional regulation, with studies promoting the exploration of utilising video games as effective tools for behavioural intervention[1].

With digital media becoming globally available and ever more prevalent in day-to-day life, it is important to ask whether the unique combination of affordances that arise from video games can be utilised in conjunction with established research in psychology to enrich existing systems of support for behavioural development.

1.2 Project Outline

This project focuses on the existent psychological research on Emotion Theory and Appraisal Theory (both distinctly linked to human motivation and behavioural patterns[3]) and explores how these theories can be drawn upon when creating an experience that enriches the player's understanding of themselves by design.

A natural concept to explore is player modelling - the creation of a 'player profile' based on the player's decisions and interactions within the game world. Data from such a player profile can be used to create more personalised gameplay which leads to a more responsive experience[4]. This project aims to explore and compare existing player typologies (structures of various factors relating to the player) by focusing on the specific factors included within each typology. By examining which factors are most prevalent, an understanding can be gained of the factors that empirically have more personal value to players.

This project includes an experimental study (designed with insight gained from background research) which asks participants to play through a short adaptive video game consisting of two parts: one of 'data gathering' which includes a short personal quiz and one of personalised gameplay. By measuring the difference against a control group, the study aims to assess the extent to which player modelling and personalisation have an effect on the capacity of the game to enact an emotional response - and subsequently foster emotional intelligence.

1.3 Goals and Scope

This project aims to achieve the following:

1. **Summarise existing literature on player modelling, emotion theory, personalisation and emotional intelligence.**

2. **Design and run a study which evaluates the impact of using player modelling and personalisation within a game when measuring self-reported improvements in emotional intelligence.**

3. **Process, present and evaluate the data using appropriate methods.**

4. **Use the data to draw insight into the extent and nature of impact which player modelling and personalisation has on the capacity of video games for behavioural intervention.**

Care has been placed to ensure that information presented within this report is clear, accurate and presented in an engaging manner with the hopes of enriching this area of research and providing insight for future research on affective games.

This chapter provides a detailed overview of existing research on the topics of player modelling, emotion theory, video game personalisation, as well as an introduction to emotional intelligence (EI). Conceptional links between the four areas will be discussed in section 3.1 (Research Aim and Conceptual Approach), providing justifications for the design choices of this study.

2.1 Player Modelling

2.1.1 Motivations and Limitations

At a high level, **player modelling** aims to provide classifications and descriptions of human players based on their characteristics, motivations, and gameplay patterns. While formal definitions have been given (for example by *Machado et al.* who define player modelling as "an abstract description of the current state of a player at a moment"[5]), it can be thought of as an umbrella term for the abstraction of player characteristics (the exact set and structure of these characteristics is often referred to as the **typology model**.)

Before discussing player modelling, it is important to highlight the limitations of any abstraction of complex human behaviour. In their overview of player typologies, *Hamari and Tuunanen*[6] highlight **two criticisms of player typologies**:

- Player types are essentially an "abstraction of an abstraction"[6]: a player typology aims to generalise player behaviours and motivations, which are abstractions of more complex thought patterns by themselves; and that
- player typologies are commonly described as discrete mutually exclusive groups when in reality psychological factors are better described in terms of scales.

Where a player typology is presented as discrete, it is important to remember that players are unlikely to fall into just one category but instead span several categories with different weights. *Hamari and Tuunanen*[6] suggest that player typing can be seen as counter-productive and reductive, commenting on the importance of player types as an aid when personalising content and not as a strict rule. Lastly, they mention that it should be questioned whether a game-specific framework is needed and if it would be more beneficial for studies to instead focus on pre-established psychological frameworks.

Still, strong motivations exist for the use of player modelling in game development and games research[7]. Player modelling can help developers design and iterate their games through an ethnographic lens, creating more personalised experiences. At the same time, researchers can take advantage of player modelling to clarify how certain motivations and aversions map onto certain types of player behaviour.

2.1.2 Overview of Existing Typology Models

This report outlines a number of existing player typologies, discussing the structure and empirical validity of each typology. Through the analysis of established literature in marketing theory, *Hamari and Tuunanen* suggest that **segmentation modes** (methods of sub-dividing a market, or in our case a player-base) can be used to categorize approaches to player modelling[6].

The four segmentation modes are Psychographic (pertaining to the person's attributes and interests), Behavioural (focusing on patterns of consumer behaviour), Geographic and Demographic. These modes are explored in depth in *Kotler and Keller's* book 'Marketing Management'[8] and offer a good overview of the attributes that can be used to describe players. It's important to not discredit demographic and geographic approaches given that emotions may vary between different digital demographics in virtual worlds - such as guilds in the MMORPG World of Warcraft [9]. While this serves as a prominent example of applied demographics within games, the majority of player modelling mainly exists within the first two modes - psychographic and behavioural. A range of notable typologies in these modes and how they have evolved is described below:

<table>
<tr><td>Bartle's Player Types (1996), Bartle R. [10]</td><td>Type: Behavioural</td></tr>
<tr><td colspan="2">Conceptual Overview
Bartle's model is based on a 2-dimensional graph representing the source of interest of players: The X-axis represents a scale of interest in other players vs. interest in the world, and the Y-axis represents a scale of 'acting with' (cooperation/exploration) vs.
'acting on' (exploitation). Within this model, each corner of the graph represents a distinct player type (Killers, Achievers, Socialisers, and Explorers), depicted by the player's sources of motivation.</td></tr>
<tr><td colspan="2">Empirical Validity and Criticisms
Though we see the model being used as the basis of later work, when examined independently the model has faced criticism of being too abstract and dichotomous. Hamari et al. propose that this stems from how the model is used rather than the original concept: "While Bartle's types are commonly used as clear-cut categories, the frameworks consist of scales instead of nominal categories..." They comment that "criticism towards Bartle's types about being too strict is partly unwarranted."[6]

In an analysis of the model, Yee found that while some types in the model showed statistical validity, the model was unsuitable as a general typology for player modelling due to the four types correlating to a high degree[11]. Despite these criticisms, Bartle's work has been used as a basis for further significant research, including by Yee himself.</td></tr>
</table>

Yee's Motivations of Play in Online Games (2007), Yee N. [11]	Type: Psychographic
Conceptual Overview Using Bartle's types as initial grounding, *Yee* transformed the model from behaviour-based to psychographic by proposing 10 sub-factors for player motivation (based on survey data collected from 3000 MMORPG players). These sub-factors were then categorised under 3 main 'umbrella' factors which motivated players: Achievement, Social, and Immersion.	
Empirical Validity and Criticisms A 2012 study by *Ducheneut, Nelson and Yee himself*[12] highlighted three criticisms of the model: it focused on the 10 sub-factors without necessarily assessing the 3 high-level factors; it was derived using a sample of English-speaking participants without accounting for differences in cultures between players; and the original results did not provide data on the empirical validity.	

Bateman et al.[13] highlight a prominent issue shared by these models: neither are designed as general, and both were intended to function only in the context of massively multiplayer games. They highlight that the existence of *Bartle's* and *Yee's* work showcases the need and motivation for a general player typology:

BrainHex (2011), *Nacke, Bateman and Mandryk [14][15]*	Type: Psychographic / Behavioural
Conceptual Overview BrainHex presents seven archetypes of players, including the Seeker (motivated by interest), Mastermind (motivated by strategy and problem-solving), and Conqueror (motivated by increasing levels of challenge). BrainHex intends to provide a model of player motivations by linking neurological understanding of players and their traits. *Nacke et al* describe the model as "an interim model – it is hypothetical in nature, and exists primarily to further the investigation of possible traits that could be used for the construction of a more robust future model"[15].	
Empirical Validity and Criticisms *Busch et al.*[16] found that though the model was found to be reliable, player scores tended to drift over time. It is also suggested that the Conqueror type might not be distinct enough from the other types. Their research shines a positive light on the reliability of BrainHex to produce player profiles, though was not able to comment on the predictive powers of BrainHex[16].	

2.1.3 Summarising the types

In a meta-synthesis on player typologies, *Hamari et al.* provide a concept-centric aggregation of twelve overarching typologies [6]. They proposed seven general categories of motivation, and found that **Sociability** and **Achievement** were the most prevalent. Bartle, Yee, BrainHex (though it wasn't included in this particular analysis) and six other works all included sociability and achievement as pivotal motivations.

Based on *Hamari et al*'s findings, we can infer that players (among many other factors) are more often driven by social interactions (community, ideas and collaboration) and by achievement (power, progress, and gaming intensity). When designing games which adapt to the player, these motivations can be utilised in conjunction with emotion theory to create affective and self-reflective experiences.

2.2 Emotion Theory and Utilisation

2.2.1 Motivations and Considerations

Emotion Theory aims to understand the nature, expression, and regulation of human emotions. There is real support and evidence for the use of video games in the space of emotion regulation (ER) and emotional development: *Bowman and Tamborini*[2] sit as prominent figures in the research of video game intervention. They provide empirical support that video games hold the potential for mood repair, more so than other types of media.

Further work has been conducted which supports this claim: *Villani et al*[1] evaluated a set of twenty-three studies pertaining to the effects of video game use for ER, concluding that when not excessive video games can be considered a valid tool for the regulation of emotions. Their concluding statement reads "The challenge for educational and psychological intervention is to exploit the emotional and affective affordances of video games, so to make use of the rich properties of these simulated experiences, as previously done with other traditional media"[1], directly supporting previous statements from *Bowman and Tamborini.*

2.2.2 Appraisal and Reappraisal for Emotion Regulation

In subsequent discussions, *Bowman and Hemenover*[17] draw on emotion theory and propose that an "analysis of video game play through appraisal and social construction models of emotions might reveal much about how video games result in emotions." This suggests that an exploration of these strategies may prove useful. *Lazarus et al.*'s work on emotion theory[18] is widely cited and defines appraisal theory as a "proposition that the **evaluation** causes the emotional response" - that emotional responses are caused by one's evaluation of themselves against their environment. Such evaluations - or 'appraisals' - produce different emotions and can occur due to numerous factors such as goal relevancy, agency and control, future expectations or factors such as self-esteem and morals[19].

Studies show that appraisal origins are external as much as they are internal: an example of this is social appraisal[20] whereby other external reactions influence our own. This suggests that appraisal theory can be used when designing personalised games which draw on such player motivations as sociability and achievement discussed in chapter 2.1.3.

The relative effectiveness of ER strategies was studied in a meta-analysis by *Hemenover and Augustine*[21]: twenty-six texts were examined pertaining to strategies such as exercise, catharsis, impression, reappraisal and distraction. They found that among strategies included in the analysis, **reappraisal** and **distraction** were most effective. Reappraisal sees participants attempt to improve their perception of an experience by focusing on the positive aspects of a negative situation, and distraction is the removal of oneself from the source of negative effect. Our study will attempt to implement above strategies when designing an affective game - such game will introduce a personalised challenge, and then provide an opportunity for reappraisal and distraction.

2.3 Personalisation

2.3.1 Motivations and Foundations

So far, this report has explored player modelling and emotion theory. Both of these concepts play a pivotal role in video game personalisation as games which are meaningful leave a stronger impact on the player. A review of player motivation by *Bakkes et al.* affirms that "involved individuals will exhibit stronger emotional reactions", citing appraisal theory as a "psychological foundation" for video game personalisation[22]. *Oliver et al.* also explore the factor of 'meaningfulness': a survey showed that meaningful games are more memorable and provide players with "feelings of added insight", reinforcing the phenomenon of personalised games enabling stronger emotional reactions[23].

2.3.2 Components of Personalisation

Bakkes et al.[22] provide a list of components within video games which can be tailored to a specific player model (or a set of player types). While not an exhaustive list, the following table provides a summary of components which will be relevant to the design of this study:

Adaptable Components: Summarisations and comments
Spaces The transformation of in-game surroundings (from various forms such as from confined to open) dependant on previous player behaviour.
Mission/Tasks The player's mission can be left open-ended, with gaps replaced during gameplay in response to the player's behaviour. This might also affect the in-game space.
Characters *Bakkes et al* talk about cases in which a complex AI would be implemented (such as first-person shooters or strategy games), they mention that it usually has to be "dumbed down" to make intentional mistakes in order to appear more realistic and that real-time adaptability (unique per experience) would require a large number of trials and hence is difficult to achieve[22]. Though *Bakkes et al* primarily talk about complex AI in games, this study will simple character adaptability (basic reactions to user decisions).
Game Mechanics The current set of game mechanics during gameplay can be adapted to the player as a form of difficulty scaling. Though the number of games that adapt their mechanics is not big, there do exist examples of this (Max Payne 3 used a secret mechanism which adjusted shooting difficulty for the player.) Since *Bakkes et al*'s review, new emotion-based approaches have been suggested, with specific applications for serious games[24].

Music/Sound
Audio adaptation is crucial for the semantic coherence of in-game worlds, but can also be used to evoke emotions and moods in the player. Before working on BrainHex, *Nacke L.*[25] contributed to a study of sonic user experience and psychophysiology in FPS games, and stated that "the creation of affective, on-the-fly sounds according to players' psychophysiological state can be used to enhance or counteract that state."[25]

2.4 Emotional Intelligence (EI)

2.4.1 Motivations and Criticisms

Though the concept of emotional reasoning as an attribute has been discussed since 1950[26], the term 'Emotional Intelligence' (or EI) became well known after the publication of *Daniel Goleman's* 'Emotional Intelligence – Why it can matter more than IQ' in 1995. In this book, *Goleman* defined EI as "the capacity for recognizing our own feelings and those of others, for motivating ourselves, and for managing emotions well in ourselves and in our relationships"[27]. In the decade following this, EI grew as an attractive commercial tool for predicting workplace performance[28] in the eyes of employers. This early commercialisation of EI (in addition to some of *Goleman*'s original claims) has sparked a lot of criticism of this construct[29]. Three major criticisms were outlined by *Cherniss C.* in 2010: the case of conflicting definitions, the need for better assessment and measurement, and the validity of EI as a predictive measurement of workspace performance[30].

In his article summarising some concerns regarding EI, *Landy FJ.* proposes a distinction between 'commercial' versus 'academic' discussions on EI. His distinction is based on the perceived ability of EI to make accurate predictions as evaluated by each group, suggesting that problems arise when the tools and set of assessed factors are blurred[31]. This gives the prospect that (when well-defined), EI holds potential as a useful tool. A recent (2020) meta-analysis has shown a link between EI and academic performance[32]. Another recent example: a 2021 diary study of emotional experience during the COVID-19 outbreak in Poland, which showed that trait EI predicted a lower frequency of negative emotions such as fear and anxiety[33]. This displays the importance of EI in our personal lives.

2.4.2 EI Models and Measurement

Literature provides three main models of EI:

<table>
<tr><td>Ability Model</td><td>Mayer JD. and Salovey P. (1997) [34]</td></tr>
<tr><td colspan="2">The ability model consists of 4 branches: emotion perception, emotion utilisation, emotion understanding and emotion management[34]. Mayer et al. have added that the model is hierarchical, with the perception of emotions as the basis of the structure and emotion management as the complex goal[35].</td></tr>
<tr><td colspan="2">Measurement and Empirical Validity
The Mayer-Salovey-Caruso Emotional Intelligence Test (MSCEIT) (V2.0) was published in 2003, self-reporting a “reasonable reliability”[36]. In a 2012 study of validity, Maul A. admits that even a decade later MSCEIT has “remained the flagship test of EI” but reported that the test suffers from a number of issues, one being the fact that the scoring for the test is based on a norm-group response, making it difficult to interpret score variation[37].</td></tr>
</table>

<table>
<tr><td>Mixed Model</td><td>Goleman D. (1998) [38]</td></tr>
<tr><td colspan="2">This model is introduced by Goleman and outlines five main ’constructs’ of EI: self-awareness, self-regulation, social skill, empathy and motivation. Within each construct, Goleman outlines a set of emotional competencies which can be learned and developed over time[38].</td></tr>
<tr><td colspan="2">Measurement and Empirical Validity
ECI (Emotional Competence Inventory) was published by Goleman, Boyatzis and Rhee(2000)[39] and though their internal reliability of the ECI scale had positive results, the publishers have allowed very few items to be peer-reviewed by external researchers[40]. For this reason, the number of peer-reviewed assessments of the ECI is limited.</td></tr>
</table>

<table>
<tr><td>Trait Model</td><td>Petrides et al. (2000, 2007) [41][42]</td></tr>
<tr><td colspan="2">The trait model views EI as a base of self-identified abilities and reflective perceptions[41]. In contrast to the above models, Petrides et al. report that the trait model does not attempt to draw a relation between EI and cognitive ability[43] - the validity of this statement has been supported since[44].</td></tr>
<tr><td colspan="2">Measurement and Empirical Validity
The Trait Emotional Intelligence Questionnaire (TEIQue) was developed by Petrides et al. to support their personality trait approach for EI[45]. In a 2016 meta-analysis, Andrei et al evaluated a pool of twenty-four studies pertaining to the validity of TEIQue[46]. Though some limitations were highlighted (most of the studies in this analysis were conducted on university students in Western countries), Andrei et al ascertain the validity of TEIQue[46]. The short-form format of the test (TEIQue-SF) has also been analysed by other studies and described as a “viable alternative to the TEIQue-LF for research in time-restricted conditions”[47].</td></tr>
</table>

Chapter 3 draws from the preceding literature and outlines the experimental aim of this study. The chapter includes a detailed design for the adaptive game tool, a discussion of a number of design challenges, and ethical considerations confronted by the study.

3.1 Research Aim and Conceptual Approach

3.1.1 Hypotheses

The nature of the research conducted by this study provides us with two hypotheses:

Null Hypothesis (H_0): The utilisation of player modelling and personalisation strategies has no effect on the capacity of a video game to evoke development of self-reported emotional intelligence.
Alternative Hypothesis (H_1): The utilisation of player modelling and personalisation strategies has statistically significant effect on the capacity of a video game to evoke development of self-reported emotional intelligence.

$\mathbf{H_0}$ is assumed initially. Depending on the results of the study, $\mathbf{H_1}$ can be accepted or rejected by running a hypothesis test on gathered quantitative data.

3.1.2 Concept Map

Chapter 2 introduced the research areas of player modelling, emotional theory, personalisation, and emotional intelligence. The below diagram demonstrates the conceptional approach which links these four areas together and which exists as the justification for this study:

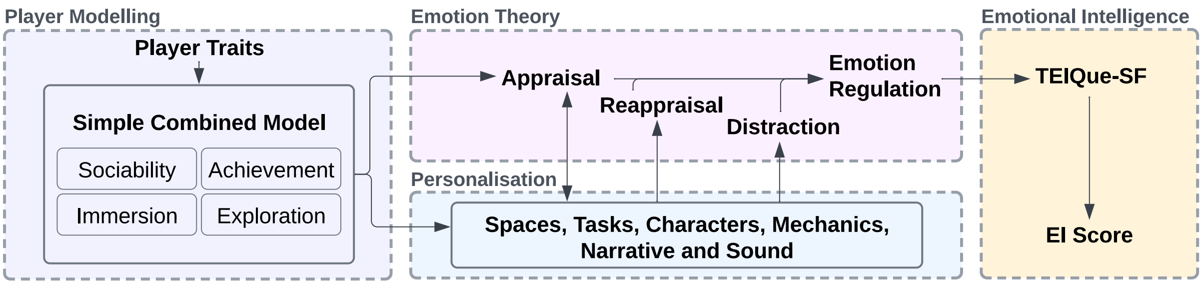

Figure 3.1: A map showing the conceptional links between player modelling, emotion theory, personalisation and emotional intelligence in respect to this study.

3.1.3 Player Modelling: Proprietary "SCM Model"

As this study is not focused on the effectiveness of any specific player modelling typology versus another, but is instead examining the impact of player modelling as a whole, it was decided the approach would not include full implementation of any existent player model. An alternative 'bite-sized' typology built from principle would be designed instead. When choosing player models for this, the validity and applications of the models had to be considered. A more general typology that can appropriately encompass a large domain of users (in a variety of game genres) was preferred. For this reason, *Bartle*'s and *Yee*'s models were not considered as they were designed to function within their own specific domains [10][11].

BrainHex was deemed an appropriate choice as a base model due to its existence as a general and "interim model"[15]. *Busch*'s study on the validity of BrainHex [16] showed strong orthogonality between the Socializer and Achiever archetypes - unlike other archetypes within the model, they have little overlap. For this reason, this study felt confident in extracting these archetypes for use in the simplified model. As explored in literature, these two traits are most prevalent in a wide array of player typologies[6], supporting the inclusion of them in our simple model. Literature on personalisation affirms that "involved individuals will exhibit stronger emotional reactions"[22]. This concept greatly ties in with the factors of immersion and exploration which are involved in a significant number of typologies as well[6].

From the abobe, the study introduces an arbitrary player model (referred to as the Simple Combined Model in Figure 3.1 and SCM in the rest of this report) composed of 4 traits-scales: **Exploration**, **Immersion**, **Sociability** and **Achievement**.

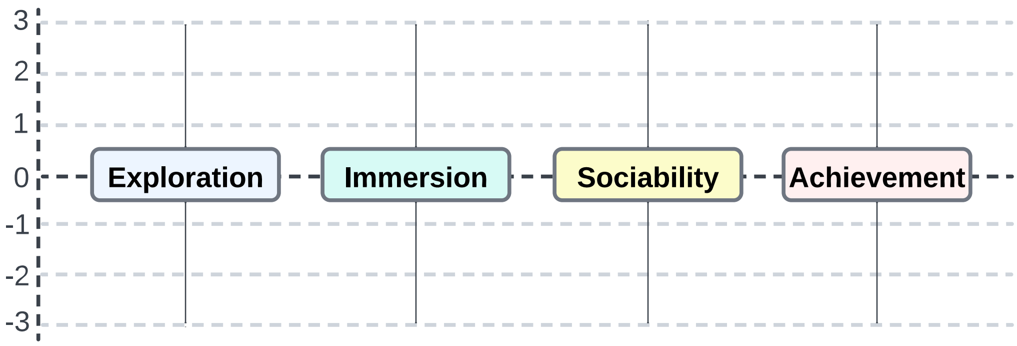

Figure 3.2: A visual representation of the SCM model - an arbitrary integer scale of -3 to 3 was chosen to maintain the simplicity of the model.

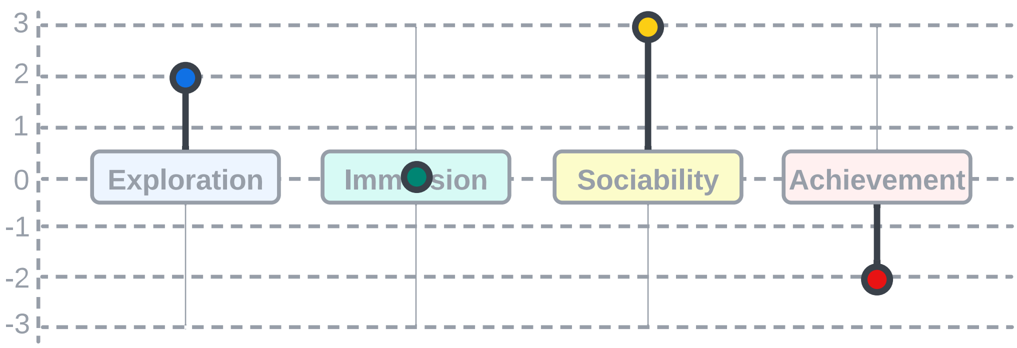

Figure 3.3: An example of a generated 'SCM Profile' in which the participant received the following scores: {Exploration:2, Immersion:0, Sociability:3, Achievement:-2}.

3.1.4 Emotion Regulation: Appraisal, Reappraisal, Distraction

Figure 3.1 depicts SCM as a base for personalisation and appraisal theory within our approach. Appraisal theory can be targeted by components of the SCM, best shown via an example: Suppose we take a random player and place them into an empty virtual room. Knowing their level of 'explorativeness', the game can be programmed to appraise exploration and curiosity if detected, improving the player's own perception of their inquisitiveness in theory.

Besides appraisal, reappraisal and distraction can be supported by certain aspects of game personalisation. A hypothetical example of personalised reappraisal: the playing of celebratory music after a very stressful/challenging level, or by having characters react positively to a successful final action (even if that action takes multiple attempts to achieve). Similar examples can be drawn when utilising personalisation for distraction - in the above, this would mean removing the initial sources of stress, the argument being that the tailoring of content around the player still counts as personalisation.

3.1.5 Emotional Intelligence: Trait EI and TEIQue-SF

The trait model of EI was chosen for this study. Unlike the ability or mixed models, the trait model is **not** designed to comment on cognitive ability. This matches the SCM and our research aims. Literature has shown that the TEIQue (a questionnaire designed to assess the trait EI of a user), as well as the short form version (TEIQue-SF), are empirically valid [46][47]. The latter was chosen as the assessment instrument for this study to reduce the total amount of time needed per participant, increasing ease of participation.

3.2 Adaptive Game Design

3.2.1 Introductory Questionnaire Room

This study aims to design and implement a game to assess the player using our SCM model and uses a generated profile to drive personalised content for emotion regulation. The game begins with an introductory room which welcomes the player, briefly explaining the controls and asking the player to complete an 'introductory questionnaire'.

The SCM model is defined by the following 'trait-scales': Exploration, Immersion, Sociability and Achievement. Four questions were designed, with the first three assessing sociability and achievement. Changes to the player's SCM profile are applied depending on the participant's answer. Please see the following question as an example:

You are working on a piece of group coursework however some members of the group are not being proactive, causing overdue tasks.		
Do you...	$\Delta_{\text{Soc.}}$	$\Delta_{\text{Ach.}}$
Try and get the members involved by booking a group meeting	+1	0
Leave it and hope that the other group members sort it out between themselves	-1	0
Take up the tasks and do them yourself to save time	-1	+1

In this example, if picking the third option for this question, the participant's sociability score will be decremented by 1, and their achievement score will be incremented by 1.
The full set of questions are included in Appendix C:

Appendix C 🗁	Full SCM Questionnaire

The final question involves using the room as a gauge for exploration and immersion:

You are placed within a boring-looking room and are told that there are multiple secrets within it. Finding those secrets does not award you anything and you have the option of skipping the room entirely and moving on.
Do you...
Skip the room
Search the room for secrets

This question aims to assess the player's curiosity. During this question, the game detects if the player explores the room they are in. If the game detects that the player has shown curiosity by exploring the room in sufficient depth, it will award them +1 $\Delta_{\text{Exp.}}$

Throughout the questionnaire the player is able to submit their answers either by pressing the numerical keys on their keyboard, or by interacting with in-game buttons corresponding to each answer. If the player chooses to not interact with the in-game buttons by the end of the quiz, they will be deducted -1 $\Delta_{\text{Imm.}}$

See Figure 3.4 for a proposed room plan which offers context for these questions:

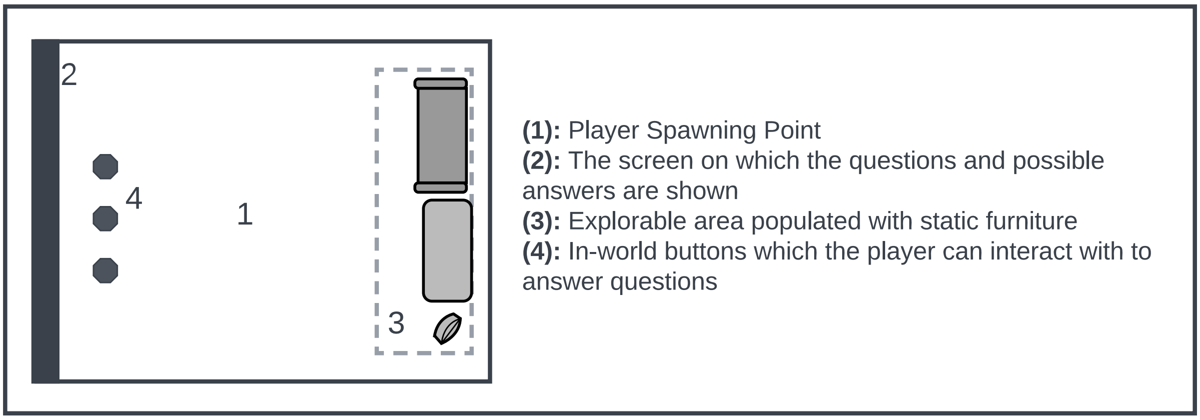

Figure 3.4: Introductory room plan

3.2.2 Exploration and Immersion

Literature (discussed in subsection 2.2.2) ascertains appraisal - including social/external appraisal - as an effective strategy for emotion regulation[21].

The game offers two opportunities for appraisal within its introduction immediately after completing the questionnaire. If the player concludes the questionnaire with an exploration score of 1 or more (i.e. they explored the room during the last question), a kind-hearted appraisal message is displayed celebrating their curiosity. Similarly, if the player concludes the questionnaire with a positive immersion score (i.e. they interacted with the in-game buttons at least once), they will be shown an appropriate appraisal message.

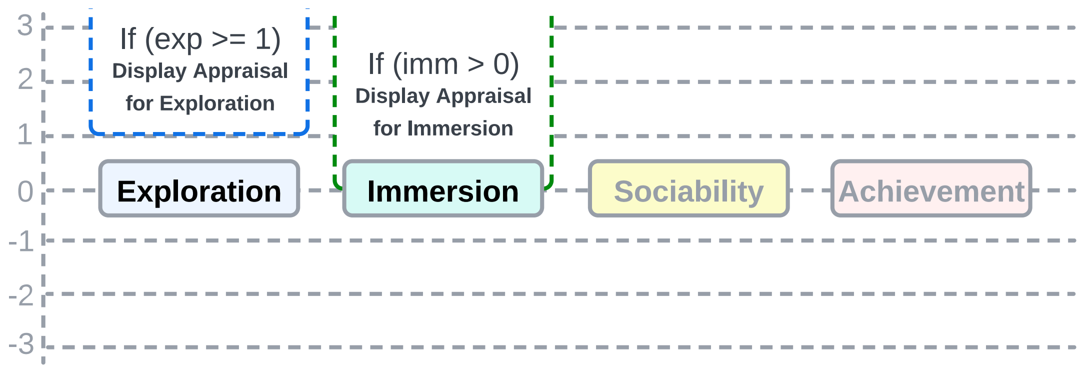

Figure 3.5: Personalisation outcomes for the exploration and immersion scales.

3.2.3 Achievement and Sociability

In addition to social appraisal, literature describes reappraisal and distraction as effective strategies for emotional regulation[21]. As a reminder, in reappraisal participants attempt to imporve their perception of an experience by focusing on positive aspects of a negative situation, and distraction is the removal of oneself from the source of negative impact.

After the completion of the introductory sequence, the study presents the player with two rooms. Each room is designed to offer a personalised challenge followed by an appropriate reappraisal. Four rooms were designed in total, two each for the SCM components of sociability and achievement:

<table>
<tr><td colspan="2">Positive Sociability Score → ’Ball-Throwing Room’ → Self-validation</td></tr>
<tr><td colspan="2">Room Description:
This room takes inspiration from the research tool ’Cyberball’ developed by Williams et al.[48][49] by having the player join a ball-throwing game in which two NPCs (non-playable characters) and the player take turns passing the ball to from one to another. After a few passes, the NPCs stop passing the ball to the player leaving them excluded. Where upbeat music was playing, it stops to reflect this atmosphere.</td></tr>
<tr><td>Challenge:
After being excluded by the NPCs, the player needs to notice that there is another ball in the room and throw that ball into a provided hoop.</td><td>Reappraisal:
Upon throwing the ball into the hoop, congratulatory music is played and a message affirms the player of their self-agency.</td></tr>
</table>

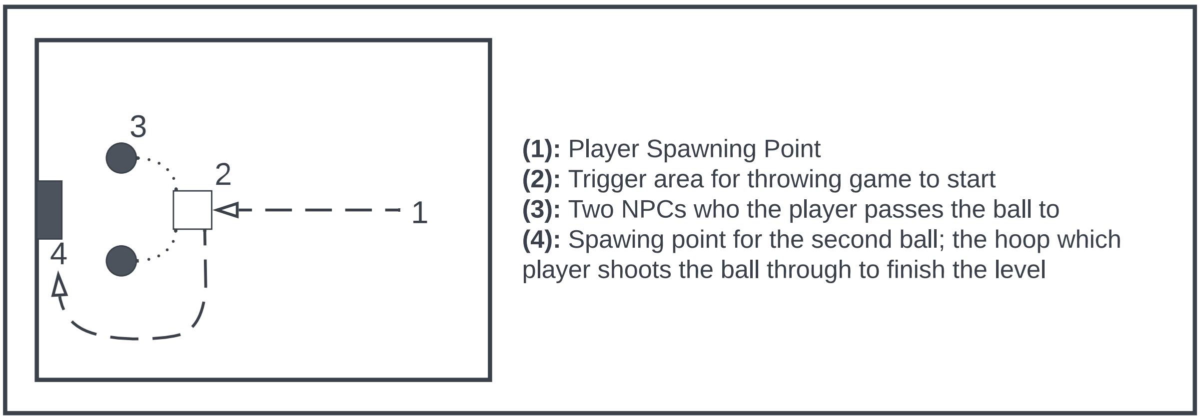

Figure 3.6: Throwing Game room plan

Negative Sociability Score → 'Jumping Game Room' → Empathy	
Room Description: In this room, players are presented with a singular NPC who appears to be periodically jumping. Players are informed of appropriate controls needed to jump.	
Challenge: This room helps the player to exercise empathy by deducing the needs of the NPC. The game requires players to match the NPC's jumping pattern (NPC provides feedback via facial expressions).	**Reappraisal:** Upon matching the correct jump patterns, the NPC is happy and the game congratulates the player.

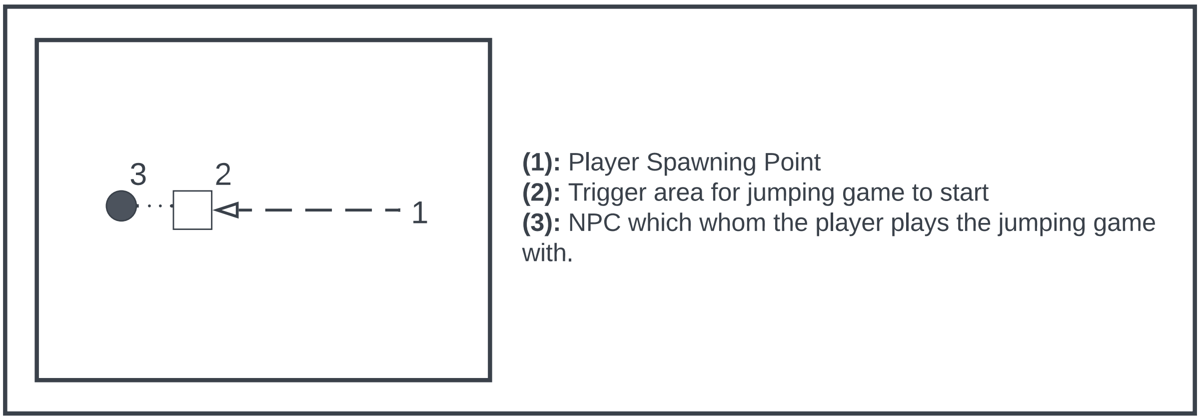

Figure 3.7: Jumping Game room plan

Positive Achievement Score → 'Pixel Art Room' → Creative Contentment	
Room Description: In this room, players are simply told to create a piece of pixel art that they are proud of. Multiple colours are offered and the player can see their artwork displayed at the end of the level.	
Challenge: This room is designed to make the player more relaxed and provide an opportunity to create something that is not assessed.	**Reappraisal:** The game provides reappraisal by thanking the player in taking their time.

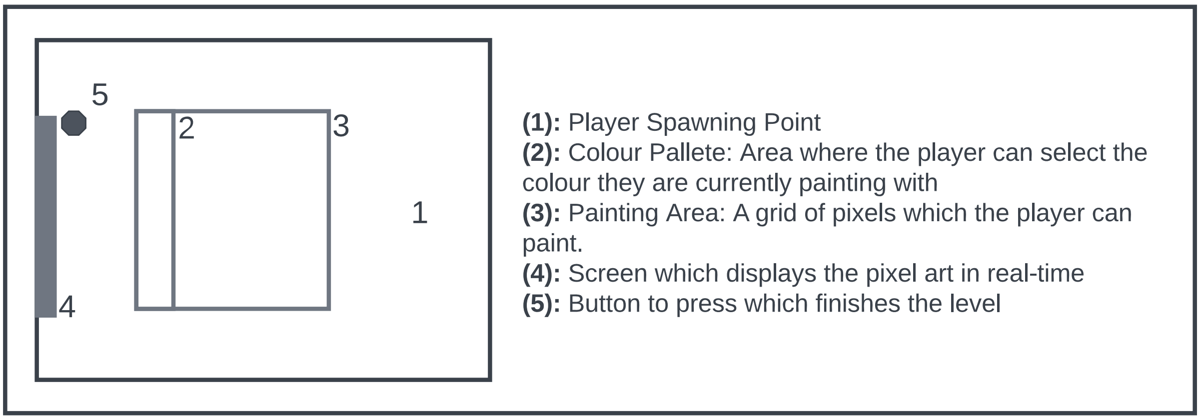

Figure 3.8: Pixel Art Room plan

Negative Achievement Score → 'Maths Quiz Room' → Eagerness to Improve	
Room Description: This room introduces an illusion of inter-participant competition by presenting a quick-fire mental maths quiz. The player is shown a goal value which they believe is the average scores of the participants before them. The player is given unlimited attempts for this challenge.	
Challenge: This room is designed to draw on and encourage the competitive side of participants.	**Reappraisal:** Upon completing the challenge, the player is congratulated.

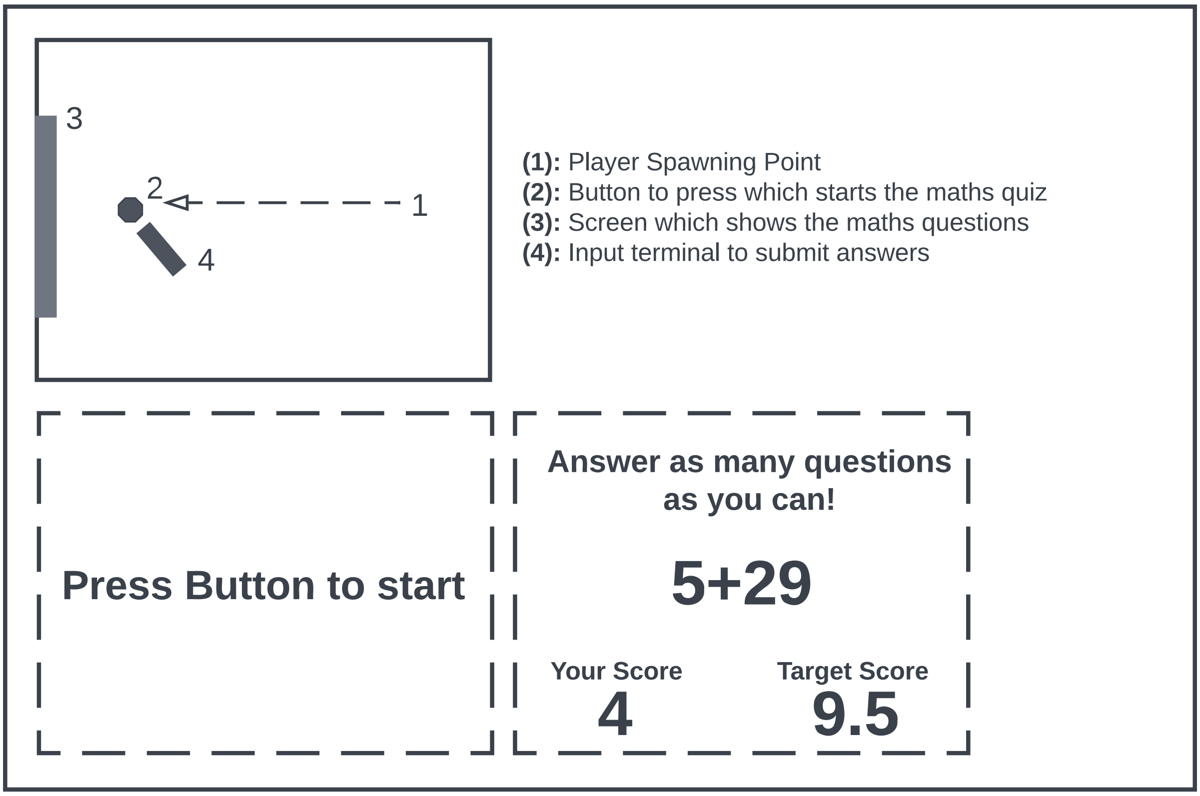

Figure 3.9: Maths Quiz Room plan and design of questions screen

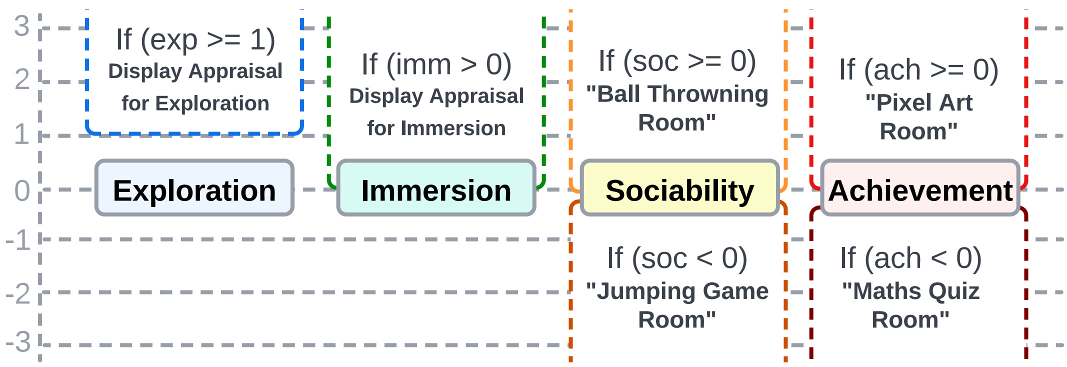

Figure 3.10: All personalisation outcomes for the four SCM scales.

3.3 Study Procedure

The below diagram outlines the comprehensive procedure for the study:

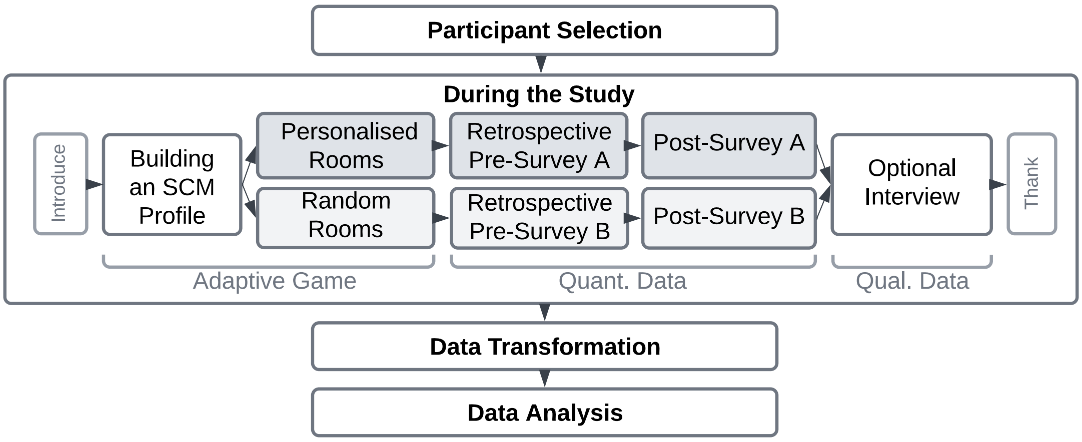

Figure 3.11: Diagram visualising the procedure of this study. The second step of the diagram is repeated for each participant. Track 'B' denotes an alternative procedure for the control group.

3.3.1 Pre-Study (Participant Selection)

The following four requirements were decided for participant selection:

1. **Current student at the University of Southampton**: This study was only open to current students at the university for ease of verification (participants could be verified as students through their university email). Having a consistent demographic sample also reduced the number of unaccounted factors in the study.

2. **A level of comfort in using conventional keyboard-and-mouse video game controls**: Though controls are explained within the application, a neuromotor familiarity with keyboard-and-mouse games is preferred for accessibility and effectiveness of participation.

3. **Stable Internet Connection**: Since this study was chosen to be remote, the participant is required to have a stable internet connection to download the application and communicate with the researcher.

4. **Computer capable of generating basic 3D graphics**: A requirement added to ensure each trial runs smoothly, as this study asks the participant to run a basic 3D game.

Though it was recognised that a large number of specific requirements reduces the possible number of participants for the study, requirements 3 and 4 had to be added in a contingency response due to challenges that arose throughout the project, described in section 6.4.

3.3.2 During the Study

The design of the game is detailed and is described in section 3.2 (Adaptive Game Design).

In order to provide two data sets that we can compare and perform a hypothesis test on, half of the participants will secretly be given a 'control group' version of the game. This version is identical to the original in every way except for the fact that it does not offer appraisal after the completion of the preliminary questions, and decides which rooms to present based on random numbers. In other words, the player's SCM profile is not utilised in the control group version of the game.

After completing the final in-game challenge, the player is thanked and presented with a link to two back-to-back surveys. The first is a **retrospective pre-survey (RPS-A)**, asking the player to answer a set of questions relating to their understanding of themselves before the study. Naturally, the second survey, the **post-survey (PS-A)**, includes the same questions but asks the participant to answer them based on their understanding of themselves after the study. As seen in Figure 3.11, a second set of surveys exists to be completed by the control group, **RPS-B** and **PS-B**. The two sets are identical with the rationale behind the separation existing only to covertly organise responses from participants in either group.

With permission from the creators at the London Psychometric Laboratory (psychometriclab.com), the 30 questions of the TEIQue-SF were utilised to produce the RPS and RP surveys for this study. The questions were rephrased to be read in past tense for the RPS, and taken verbatim for the PS. Copies of any measurement tools are included in the archive for the interest of the reader:

Appendix A 🗁	Archive Table of Contents, including the RPS, PS, and original TEIQue-SF measurement tool.
Appendix D 🗁	Permission to adapt the TEIQue-SF.

After submitting the surveys, participants are invited to complete an optional short semi-structured interview. The purpose of the interviews is to offer an insight on the effectiveness of the study from the participant's perspective which can be used to supplement the quantitative data from the surveys. Responses during the interview may also provide insight into the limitations of the study that can be used to enrich the quality of future work in this field. The interview consists of six questions:

1. **What were your experiences and thoughts during the starting quiz of the game?**
   Allows the participant to comment on the quiz which is used to build their SCM profile, including their opinions on the questions and the environment in which the questions are asked.

2. **What rooms stood out to you the most, and why?**
   Allows the participant to comment on the challenges which the game presented to them. This can give a good insight into which rooms were more memorable or effective.

3. **What were your initial reactions when you met the in-game characters?**
   Some of the challenge rooms include interacting with non-playable characters (see section 3.2 (Adaptive Game Design)). This questions allows the participant to describe what they thought of the characters and how the characters made the participant feel.

4. **Do you feel like your answers to the quiz at the start affected your experience in the rest of the game?**
   Allows the participant to share their perception of any personalisation in the game. This will allow us to analyse the effectiveness of the study's design by comparing answers between the experimental and control groups.

5. **How has playing through the game and reflecting on your experience made you feel right now?**
   To accept $\mathbf{H_1}$, the study needs to provide sufficient statistical evidence of self-reported EI improvement - responses to this questions might help to provide explanation for insufficient statistical evidence or support them.

6. **Do you think there is anything important that I did not ask or anything else you would like to share?**
   This open questions allows participants to share any other comments which may be useful for this study.

A semi-structured format was chosen for the interview to ensure a consistency of structure between trials while still allowing a deep richness of responses. The questions were left open, short, and did not contain technical language to encourage response depth.

3.3.3 Post-Study (Data Transformation and Analysis)

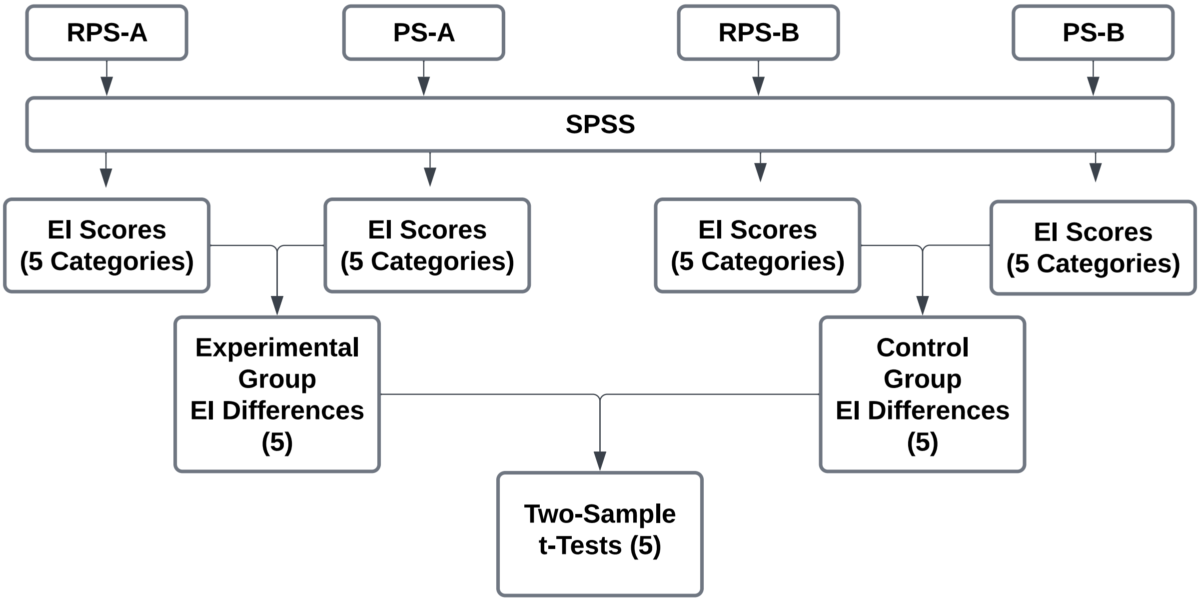

Figure 3.12: Figure showing the data flow when processing the quantitative survey response data.

Quantitative data (numerical data from survey responses) will be processed following these steps:

1. Data from surveys is gathered and downloaded. Each survey response yields a set of 30 integer values (one for each question), each ranging from 1 to 7 (denoting the 7-point Likert scale response to each question).

2. Each set of values is processed using IBM's SPSS Software using a script provided by the London Psychometric Laboratory[50]. This script is provided as the below appendix:

Appendix E 🗁	SPSS Script provided by London Psychometric Lab

   The script generates 5 factor values (well-being, self-control, emotionality, sociability and global trait EI) for each survey.

3. Differences between all of the values are calculated, yielding two sets of differences (for experimental group and control group).

4. A Two-Sample t-Test between the experimental group differences and the control group differences is performed for each factor (5 tests in total)

The results of the five tests will allow us to comment on the changes in individual factors, as well as general trait EI (this will be the final test and will be used to accept/reject $\mathbf{H_1}$). Two-Sample t-Tests were chosen as a statistical tool based on the following conditions as per guidance from the article 'Hypothesis Tests Explained' by Towards Data Science[51]: Trait EI scores are assumed to be normally distributed; we are testing more than one sample; we are performing a test to see the difference between the two sets; and the two data sets are unpaired (they are independent). See Figure 3.13 for a flowchart supplied by Towards Data Science as guidance for selecting statistical tests:

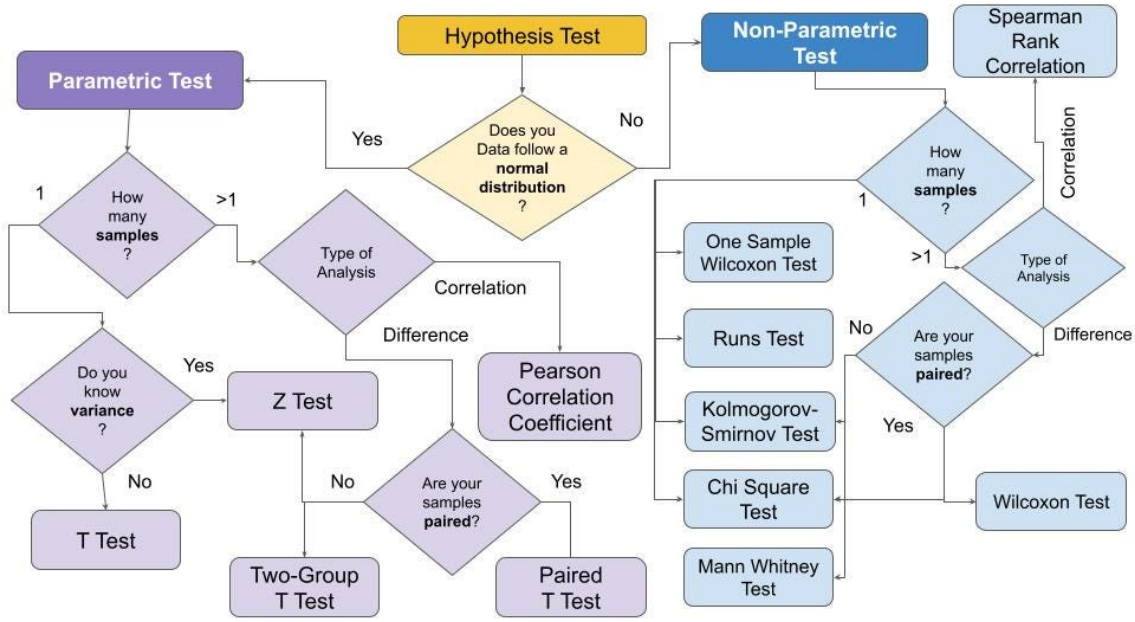

Figure 3.13: Flowchart supplied by Towards Data Science as guidance for selecting statistical tests [51]. Following this flowchart, Two-Sample t-Tests were chosen for this study.

The final quantitative results will be triangulated with qualitative interview responses in order to add additional context and content for discussion. To enable this, interview responses will undergo a thematic analysis in which key themes will be highlighted and discussed.

3.4 Addressing Biases

The study strives to acknowledge and addresses common biases that might arise during the procedure:

Response-Shift Bias:
Described by *Howard GS*, Response-shift bias occurs when the intervention changes the participant's frame of reference when responding[52]. For this study, this might imply that participants rate themselves higher on the second survey because they know the context of the study, and not because the game has benefited them. In an attempt to limit the effects of this bias, the procedure introduces both surveys after the game (known as a retrospective pre-test/post-test method) as per guidance from *Sibthorp et al*[53].

Observation Bias and **Social Desirability Bias**:
To limit the presence of observation bias and social desirability biases, all data gathered from the surveys will be fully anonymised and the researcher will not be present during the trial with exception to conduct the optional interview or to answer any questions throughout the study.

Performance Bias:
Performance bias is addressed well within this study. Though a participant might rightly assume the existence of two groups in the study, a strong care has been made to not reveal this information. Both groups complete the same questionnaire, survey, and interview questions and no external changes are presented within the game that might allude the player to their assigned group.

Confirmation Bias:
To minimise the effects of confirmation bias that might stem from the researcher's beliefs, all quantitative data gathered is analysed using rigorous statistical methods. Any conclusions which may arise from the qualitative data will be assesses and presented, including those which support the rejection of $\mathbf{H_1}$.

Question-Order Bias:
To minimise question-order bias, the surveys will be administered using software which has 'random question ordering' functionality.

Inconsistency:
Inconsistency is reduced throughout the procedure by not in any way altering any data gathering tools during the entire duration of the study and by only asking questions set out out in the interview plan.

3.5 Ethics Approval

An application detailing all methodology and data protection practices was submitted to the University of Southampton's Faculty of Engineering and Physical Sciences Ethics Committee.
Including subsequent amendments, the application was approved on 12th April 2023 under reference 78988.A1
All involved participants were presented with an information sheet and have signed explicit consent. Relevant ethics and data protection documents are provided within the archive for review:

Appendix A 🗁	Archive Table of Contents, including Participant Information Sheet, Consent Form, and FEPS Data Protection Plan.

In the context of this report, **'study goals'** are defined as criteria to be met for the study to fulfilling its ambitions: **These goals are not limited to technology (adaptive game); they span the entire study procedure and will be used to evaluate the project in chapter 8.**

4.1 Review of Goals

As set out in section 1.3, the study outlines 4 main goals:

- **G1**: **Summarise existing literature.**
- **G2**: **Design and run a study to gather quantitative and qualitative data which can be used to accept/reject H_1 .**
- **G3**: **Process, present and evaluate the data using appropriate methods.**
- **G4**: **Use the data to accept/reject H_1 and draw a conclusion.**

4.2 Technical Requirements

Technical requirements are separate from the study requirements and are defined below with MoSCoW prioritisation. These requirements will be used to evaluate the completeness of the adaptive game and surveys:

T1: **The game and surveys should be complete and fit the purpose of the study**		
Base Functionality		
T1-REQ1	The game must run on a participant's machine.	**MUST**
T1-REQ2	The game's controls must be usable and conventional.	**SHOULD**
T1-REQ3	The game should not be cumbersome to download and run.	**SHOULD**
Introductory Questionnaire		
T1-REQ4	The game should be able to display questions and available answers.	**MUST**
T1-REQ5	The player can interact with in-world buttons or press the numerical keys on their keyboard to answer questions.	**MUST**
T1-REQ6	Appropriate messages should display when appropriate for exploration and immersion appraisal.	**MUST**
T1-REQ7	The game should build a player profile accurately and use it for personalisation.	**MUST**
Rooms for Emotion Regulation		
T1-REQ8	Core aspects of the 'Ball Throwing Game Room' function as intended.	**MUST**

T1-REQ9	Core aspects of the ’Jumping Game Room’ function as intended.	**MUST**
T1-REQ10	Core aspects of the ’Pixel Art Room’ function as intended.	**MUST**
T1-REQ11	Core aspects of the ’Maths Quiz Room’ function as intended.	**MUST**
T1-REQ12	**All** aspects of the ’Ball Throwing Game Room’ function as intended.	**SHOULD**
T1-REQ13	**All** aspects of the ’Jumping Game Room’ function as intended.	**SHOULD**
T1-REQ14	**All** aspects of the ’Pixel Art Room’ function as intended.	**SHOULD**
T1-REQ15	**All** aspects of the ’Maths Quiz Room’ function as intended.	**SHOULD**
Surveys		
T1-REQ16	Retrospective Pre-Survey responses collected without error.	**MUST**
T1-REQ17	Post-Survey responses collected without error.	**MUST**
T1-REQ18	Game directs participants to the survey at the end.	**COULD**

5.1 Tools

- **The Unity engine** was chosen as the primary development tool due to cross-platform building (enabling participation for Windows, Linux and Mac users). In addition to this, by leveraging on my familiarity with the engine I was able to save time by not needing to learn new tools. Version 2021.3.9f1 was chosen as the latest available LTS (Long Term Support) release of the engine, ensuring a stable development environment for the duration of the project. C# scripting allows for fast prototyping and separation of concerns throughout development.

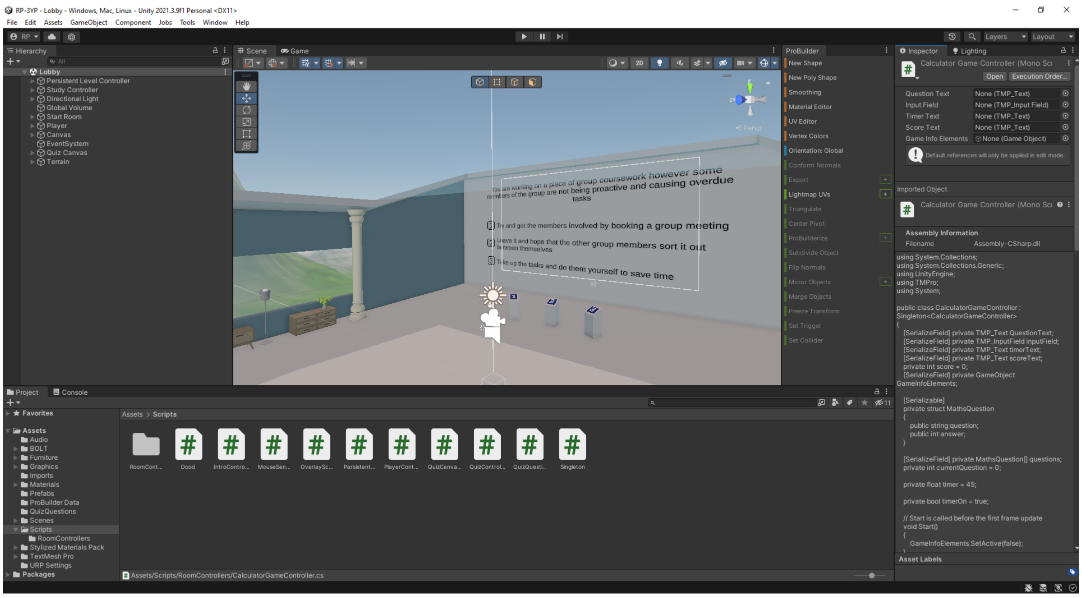

Figure 5.1: The Unity Editor GUI. As seen at the bottom, many C# scripts can be written to provide logic for in-game objects.

- **Visual Studio** was chosen as the IDE for these scripts, primarily for its notable integration with Unity (automatic features such as auto-completion and API lookup).

- **Scriptable Objects** are definable data containers which can enable quick creation of reusable assets from a programmer-defined template. Within this project, scriptable objects were used to define questions within the introductory questionnaire (see Figure 5.2).

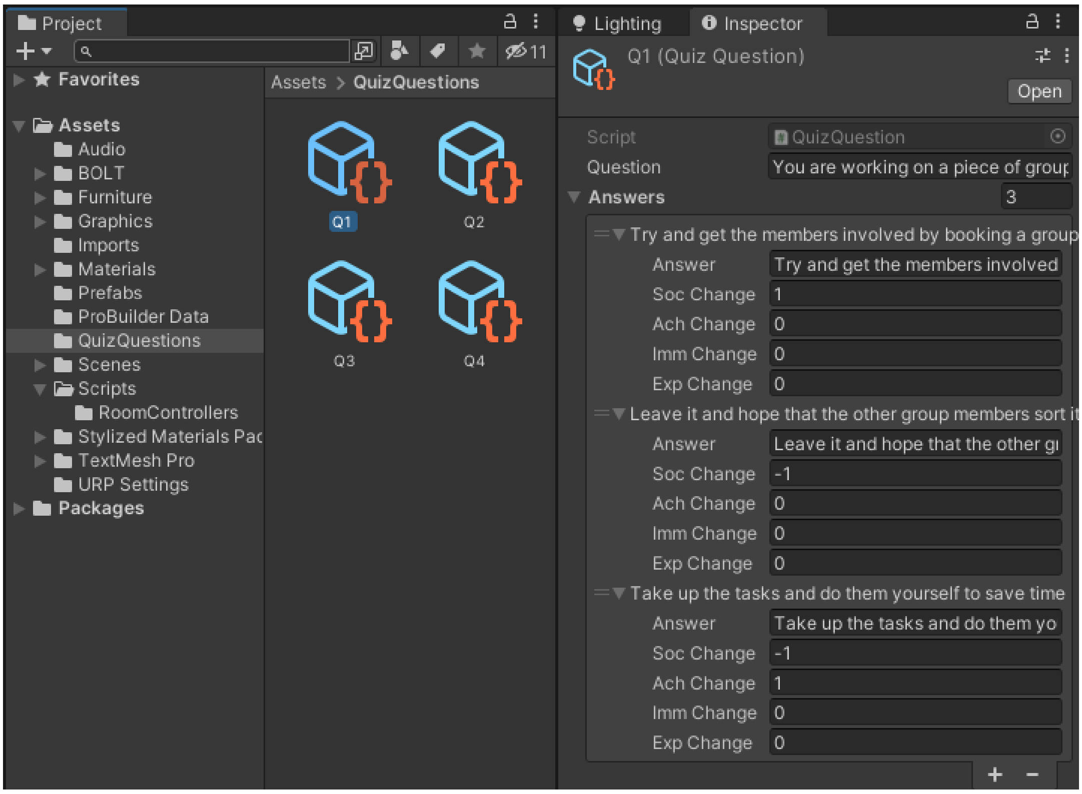

Figure 5.2: Shows four programmer-defined 'Quiz Question' scriptable objects. Each object acts as a structured container for data such as the question string, answers and the effects of the answers on the SCM profile.

- **ProBuilder** is a 3D level prototyping tool integrated into Unity which allows for the creation of levels without the need of external tools such as Blender). All of the rooms within the game were modelled using ProBuilder (see Figure 5.3).

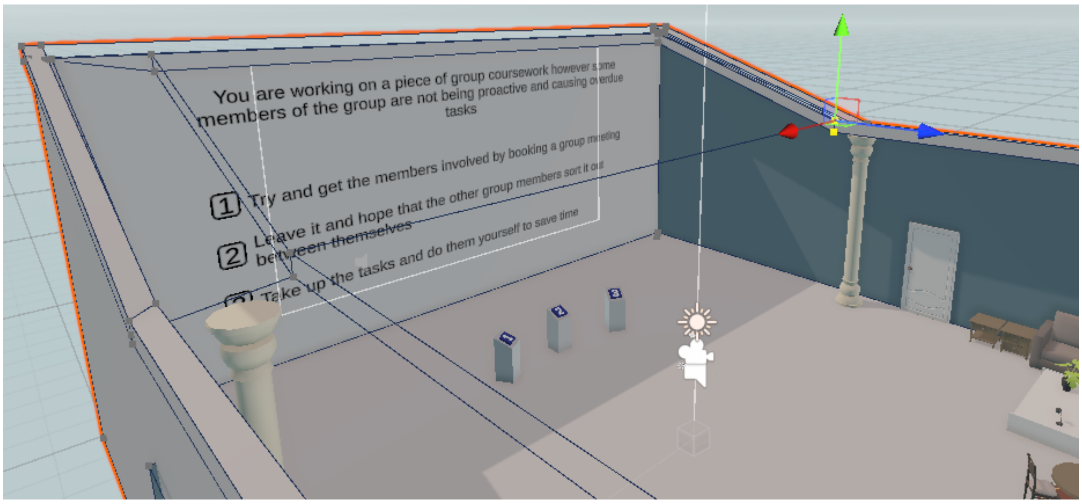

Figure 5.3: Probuilder was used to model all of the rooms within the game through manipulation of vertices, edges and faces.

5.2 Additions During Development

There exist a few minor differences between the design outlined in section 3.2 (Adaptive Game Design) and the final implementation. These are discussed below:

Change 1: Informational Tooltips
Introduction tooltips were added at the start of the game and at the beginning of each level (Figure 5.4). By including all relevant information within the level, the game becomes self-contained adding to the autonomy of each trial (hence helping to reduce observation bias by minimising the amount of information exchange needed between the participant and researcher).

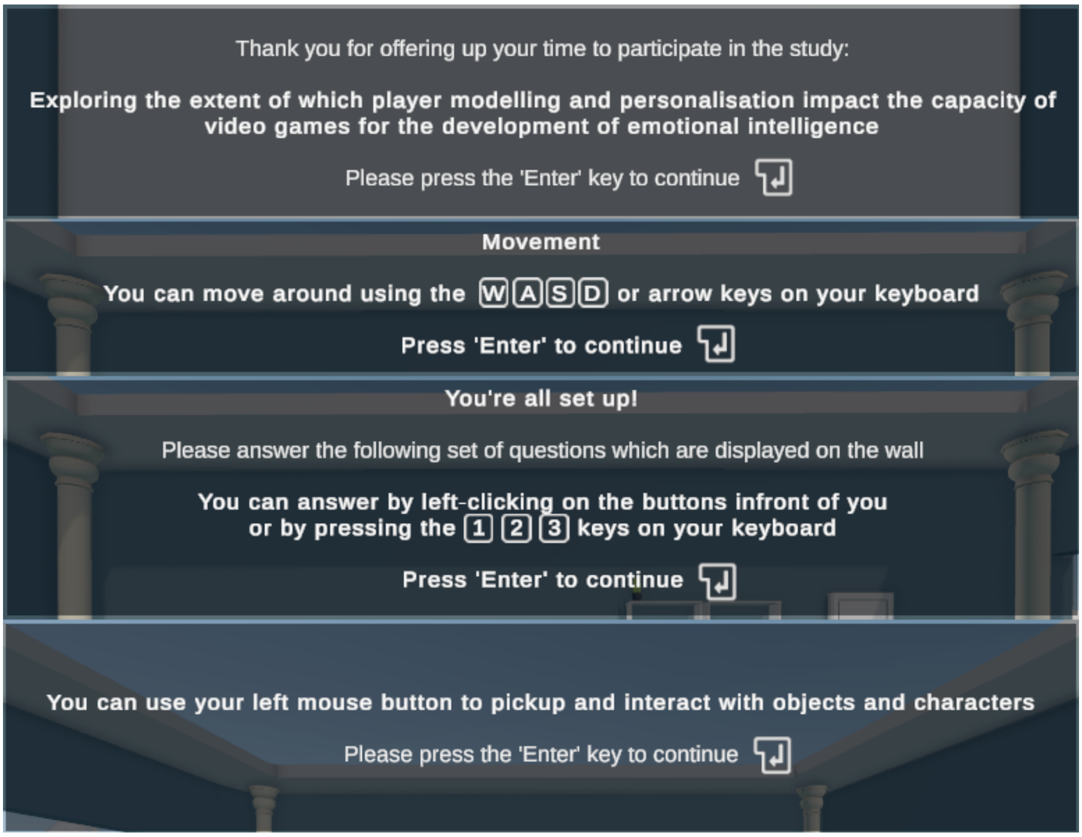

Figure 5.4: Examples of new tooltips (non-exhaustive) which are shown to the player when they launch the game or enter a new room.

Change 2: Mouse Sensitivity Calibration
Mouse sensitivity calibration was added during implementation, allowing the user to change the sensitivity of their in-game aim at any time during the trial (Figure 5.5). This feature makes the game more accommodating and comfortable to play.

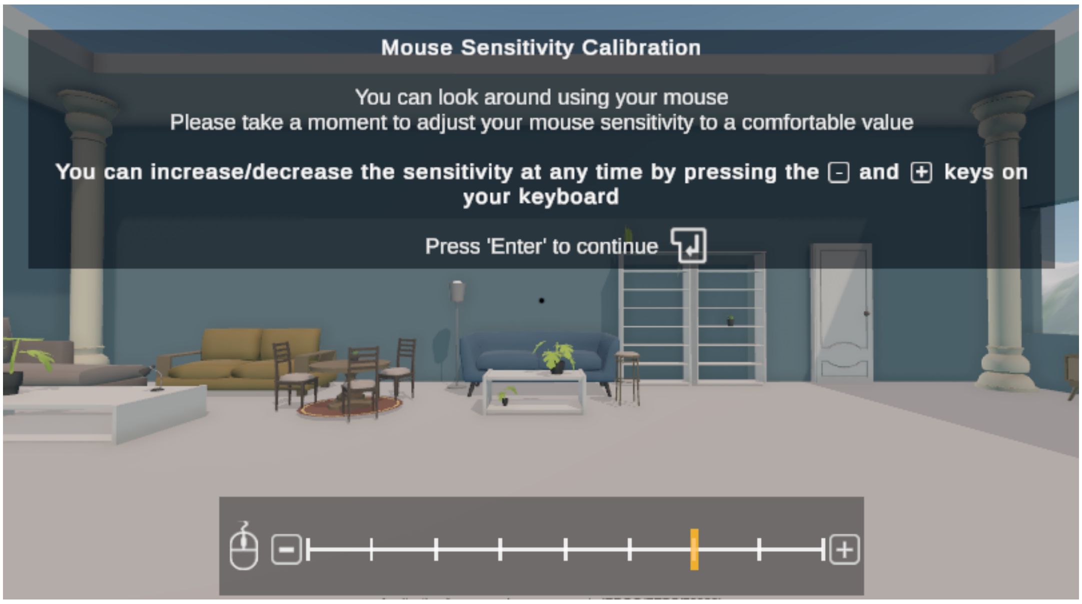

Figure 5.5: New mouse sensitivity calibration feature.

5.3 Game Testing

5.3.1 Testing Strategy

A mix of strategies were employed for this project:

- For controls: unit testing was facilitated by using Unity's Test Framework (UTF) - an integration of NUnit.
- For qualitative and non-game related requirements: written justifications and visual evidence are provided.

5.3.2 Technical Evaluation

Base Functionality		
T1-REQ1	The game must run on a participant's machine.	Partially Met
T1-REQ2	The game's controls must be usable and conventional.	Met
T1-REQ3	The game should not be cumbersome to download and run.	Partially Met

T1-REQ1: As per Unity's minimum requirements, a machine must have a capable GPU in order to run games made in Unity. In order to remedy this, an additional requirement was added to the participant selection stating that the participant must have a computer capable of generating basic 3D graphics. This requirement was partially met.

T1-REQ2: The game uses conventional WASD and mouse controls. To ensure usability, the game is also controllable with arrow keys. A set of 16 unit tests were written to emulate user input and automatically test orthogonal and combinatorial player movement (Figure 5.6). All 16 tests pass, evaluating **T1-REQ2** as 'Met'.

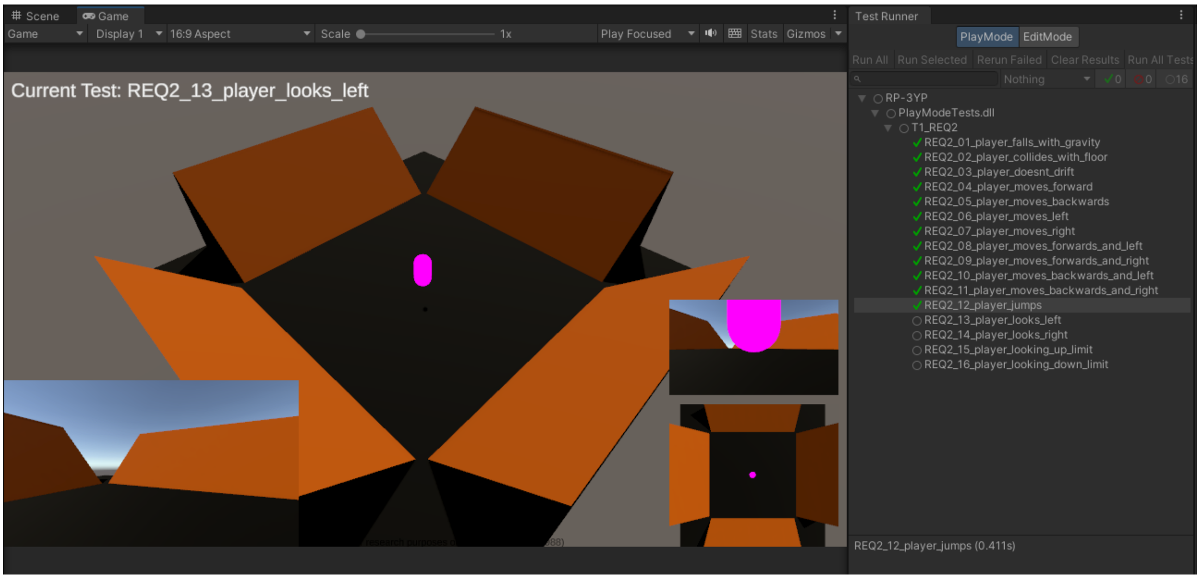

Figure 5.6: A testing environment was created and a set of 16 unit tests were written testing the movement of the player (including looking around with the camera). The figure depicts a point midway through the test sequence.

T1-REQ3: When zipped, the total size of the all necessary files needed to run the game was 64Mb. Participants are required to have a stable internet connection as per the participant selection requirements, however this still might prove to be a barrier during the study and so was evaluated as 'partially met'.

Introductory Questionnaire		
T1-REQ4	The game should be able to display questions and available answers.	Met
T1-REQ5	The player can interact with in-world buttons or press the numerical keys on their keyboard to answer questions.	Met
T1-REQ6	Appropriate messages should display when appropriate for exploration and immersion appraisal.	Met
T1-REQ7	The game should build a player profile accurately and use it for personalisation.	Met

T1-REQ4, **T1-REQ5**: Evidenced in Figure 5.7

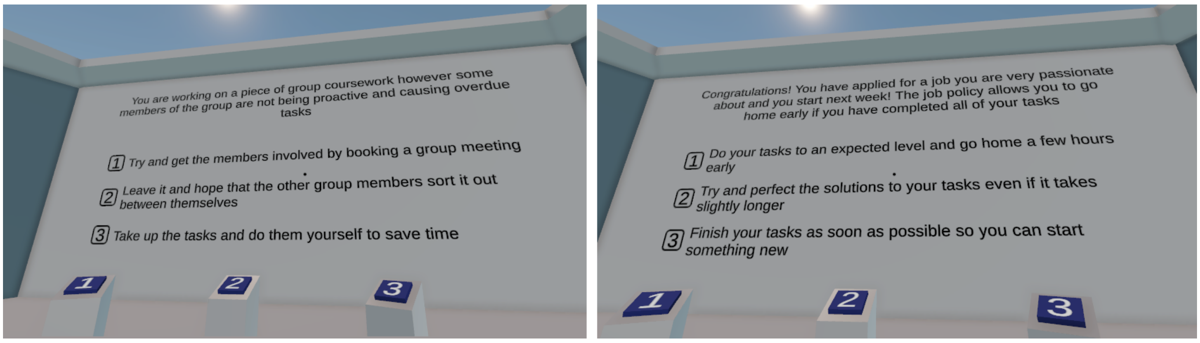

Figure 5.7: Questions and answers on the screen change after interacting with a button.

T1-REQ6: Evidenced in Figure 5.8

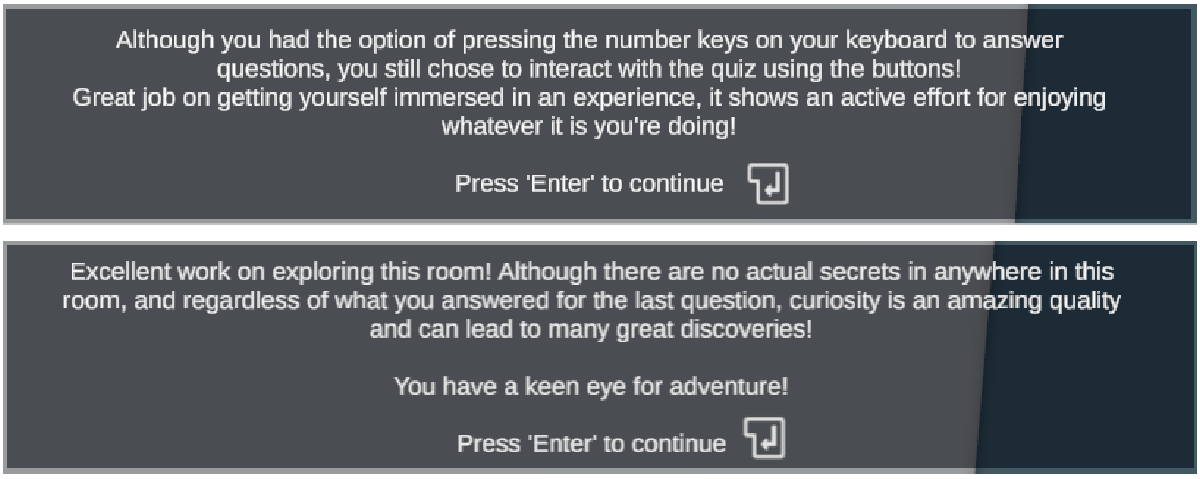

Figure 5.8: The above appraisals are shown when the player interacts with the in-game buttons at least once or explores the room during the final question of the questionnaire.

T1-REQ7: Evidenced in Figure 5.9

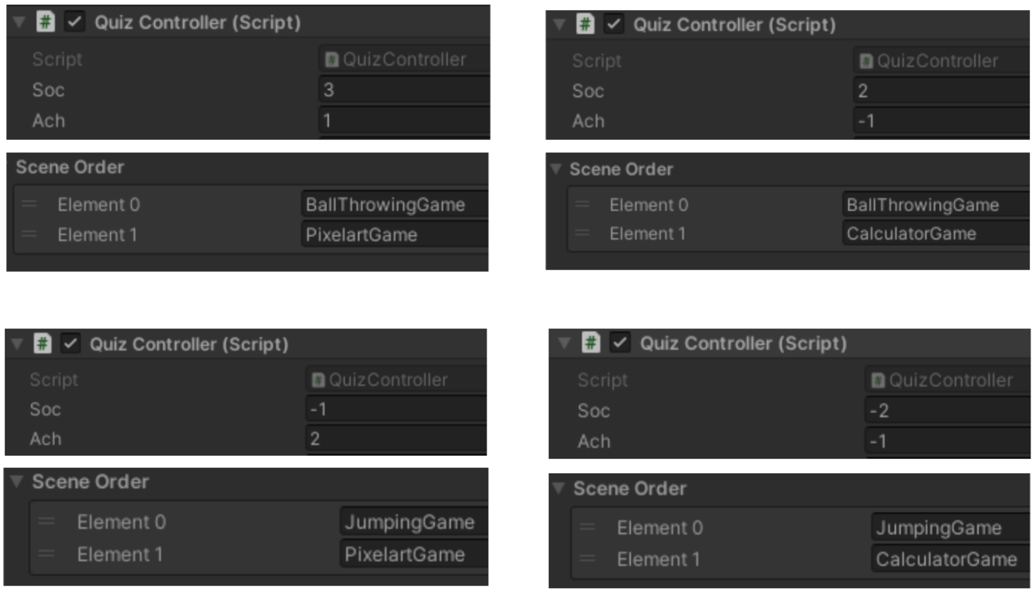

Figure 5.9: Testing the four possible room combinations.

Rooms for Emotion Regulation		
T1-REQ8	Core aspects of the 'Ball Throwing Game Room' function as intended.	Met
T1-REQ9	Core aspects of the 'Jumping Game Room' function as intended.	Met
T1-REQ10	Core aspects of the 'Pixel Art Room' function as intended.	Met
T1-REQ11	Core aspects of the 'Maths Quiz Room' function as intended.	Met
T1-REQ12	**All** aspects of the 'Ball Throwing Game Room' function as intended.	Met
T1-REQ13	**All** aspects of the 'Jumping Game Room' function as intended.	Met
T1-REQ14	**All** aspects of the 'Pixel Art Room' function as intended.	Met
T1-REQ15	**All** aspects of the 'Maths Quiz Room' function as intended.	Met

T1-REQ8, **T1-REQ12**: Evidenced in Figure 5.10.

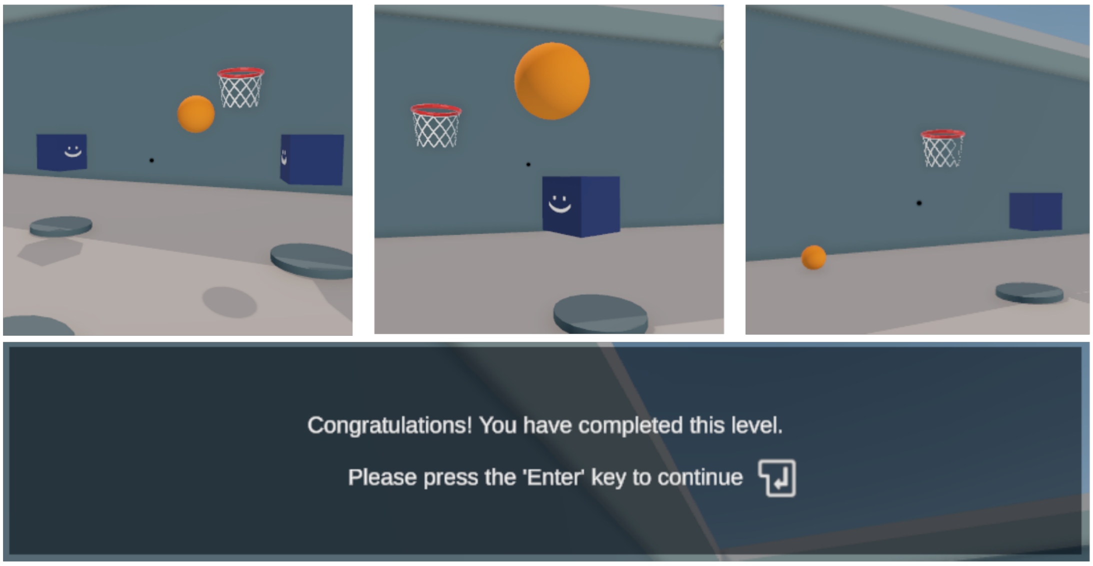

Figure 5.10: Ball Throwing Game functioning as intended: NPCs begin throwing the ball to the player who is able to throw it back. After a few throws, the NPCs stop including the player and the music stops. After 15 seconds, a new ball appears and the player is able to throw it into the hoop to complete the level.

T1-REQ9, **T1-REQ13**: Evidenced in Figure 5.11.

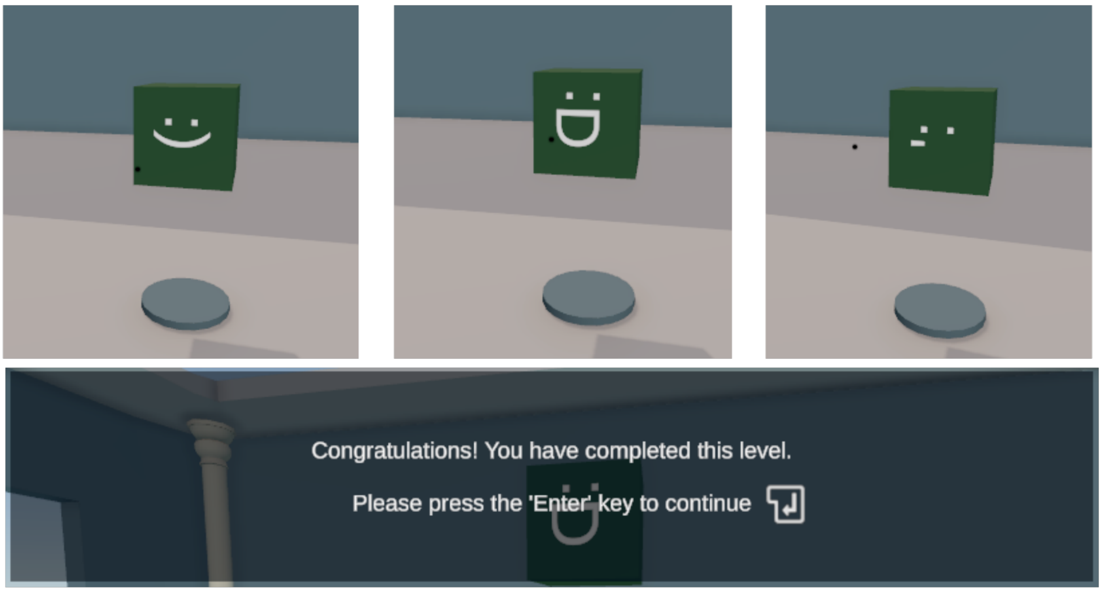

Figure 5.11: Jumping Game functioning as intended: The NPC jumps and changes facial expressions depending on how well the player is matching the jumps.

T1-REQ10, **T1-REQ14**: Evidenced in Figure 5.12.

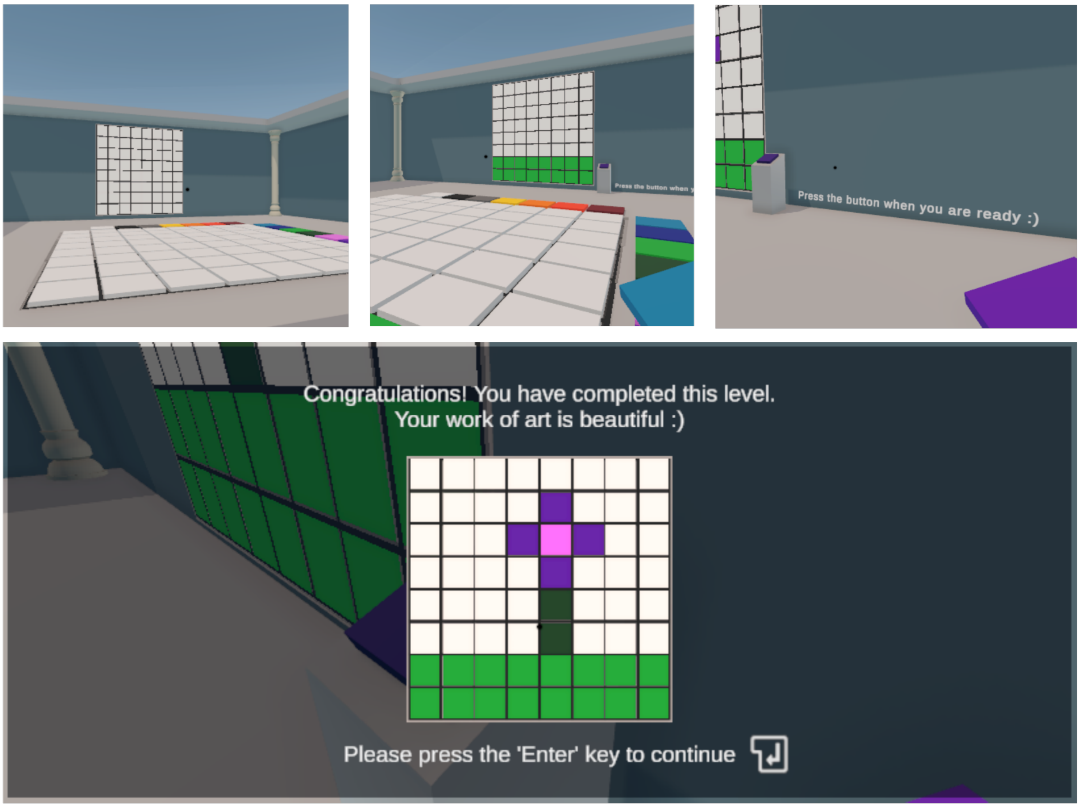

Figure 5.12: Pixel Art Game functioning as intended: The player is able to choose from various coloured tiles and place them down. The artwork is updated on the wall in real time and is displayed when the player finishes the level.

T1-REQ11, **T1-REQ15**: Evidenced in Figure 5.13.

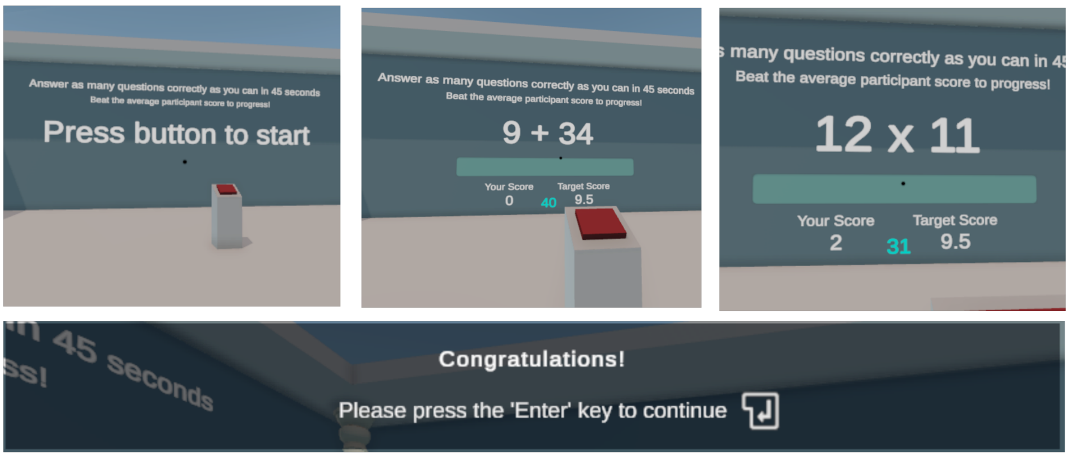

Figure 5.13: Maths Quiz Game functioning as intended: The player is able to press the button to start and they can type their answers to displayed questions. Score counters and countdown work as intended and the level is completed once a score of 10 is reached.

Surveys		
T1-REQ15	Retrospective Pre-Survey responses collected without error.	Met
T1-REQ16	Post-Survey responses collected without error.	Met
T1-REQ17	Game directs participants to the survey at the end.	Met

T1-REQ15, **T1-REQ16**: Surveys can be completed and responses are downloadable in Excel format from iSurvey.

T1-REQ17: Evidenced in Figure 5.14

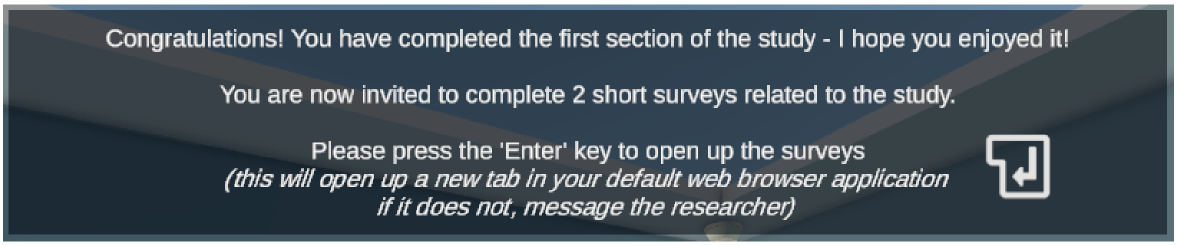

Figure 5.14: A tooltip appears at the end of the second level. Pressing enter opens the survey link in the default browser.

6.1 Skills Audit

I conducted the following skills audit on myself in early January to determine areas of needed improvement:

General Skills	
Skill	Score (1-5)
Research Skills (Reading and Interpreting)	4
Game Design and Development	4
Statistics and Data Analysis	2
Time Management	3
Technical Skills	
Unity and C#	5
3D Modelling	3

Following from this, I have decided to invest time into learning about statistical analysis and hypothesis testing.

6.2 Risk Assessment

Problem	Loss	Prob.	Risk	Mitigation
Laptop is lost or damaged	5	2	10	All work to be saved on the cloud (Overleaf, OneDrive, GitLab)
Failure to stay up to date with schedule	4	4	16	Allow more time for completion of tasks than initially thought
Inability to develop the game in time OR Failure to secure ERGO approval in time	5	2	10	Be knowledgeable in literature to be able to adapt project quickly
Inability to use SPSS Software	3	4	12	Perform manual calculations or utilise London Psychometric Lab's online scorer
Not enough participants sign up	4	3	12	Start participant selection early

6.3 Project Timeline

6.3.1 Original Schedule

Figure 6.1 outlines the original schedule for the project. This schedule aimed to reduce workload during deadline periods and assumed two iterations of ERGO approval.

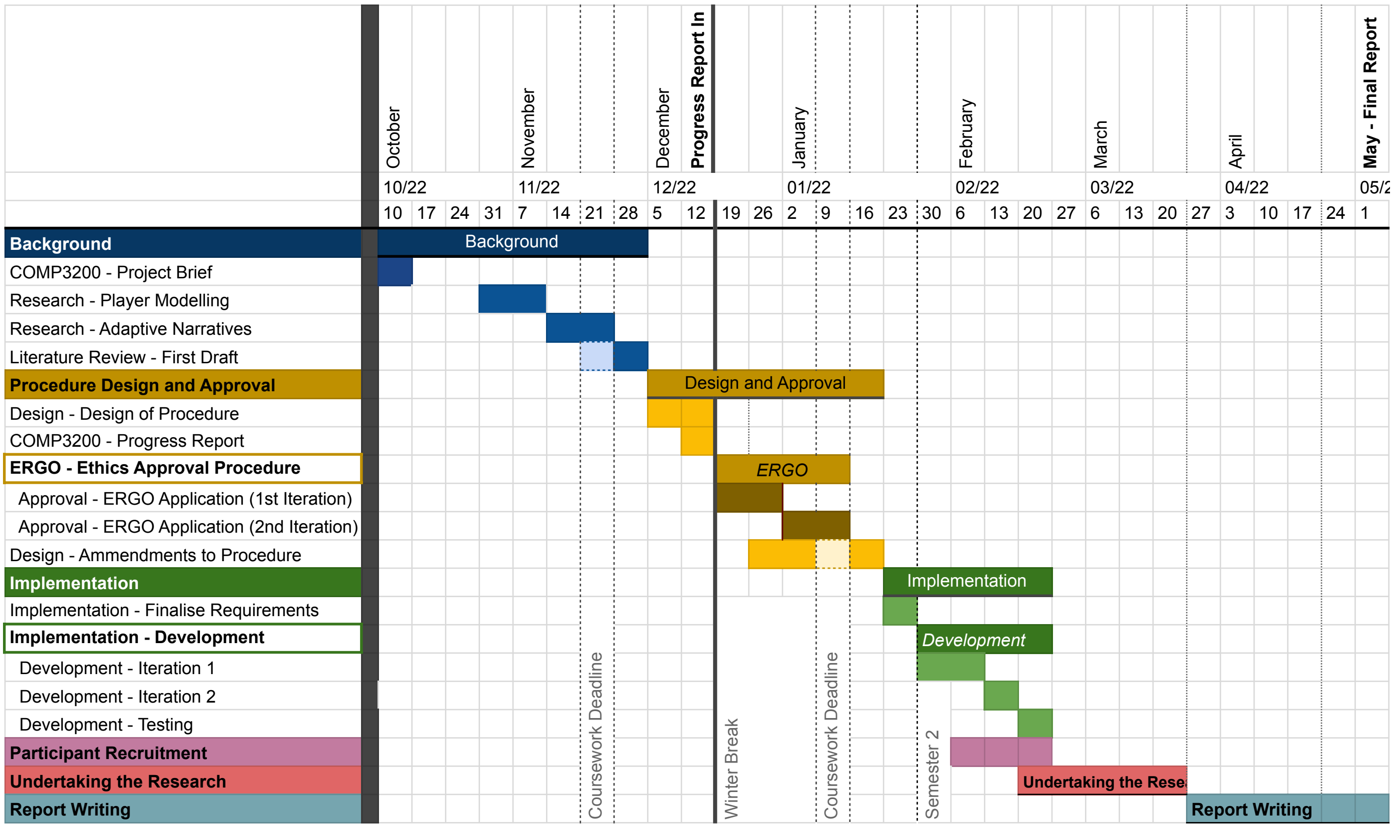

Figure 6.1: Original Gantt chart as set out in December (progress report handin).

6.3.2 Final Schedule

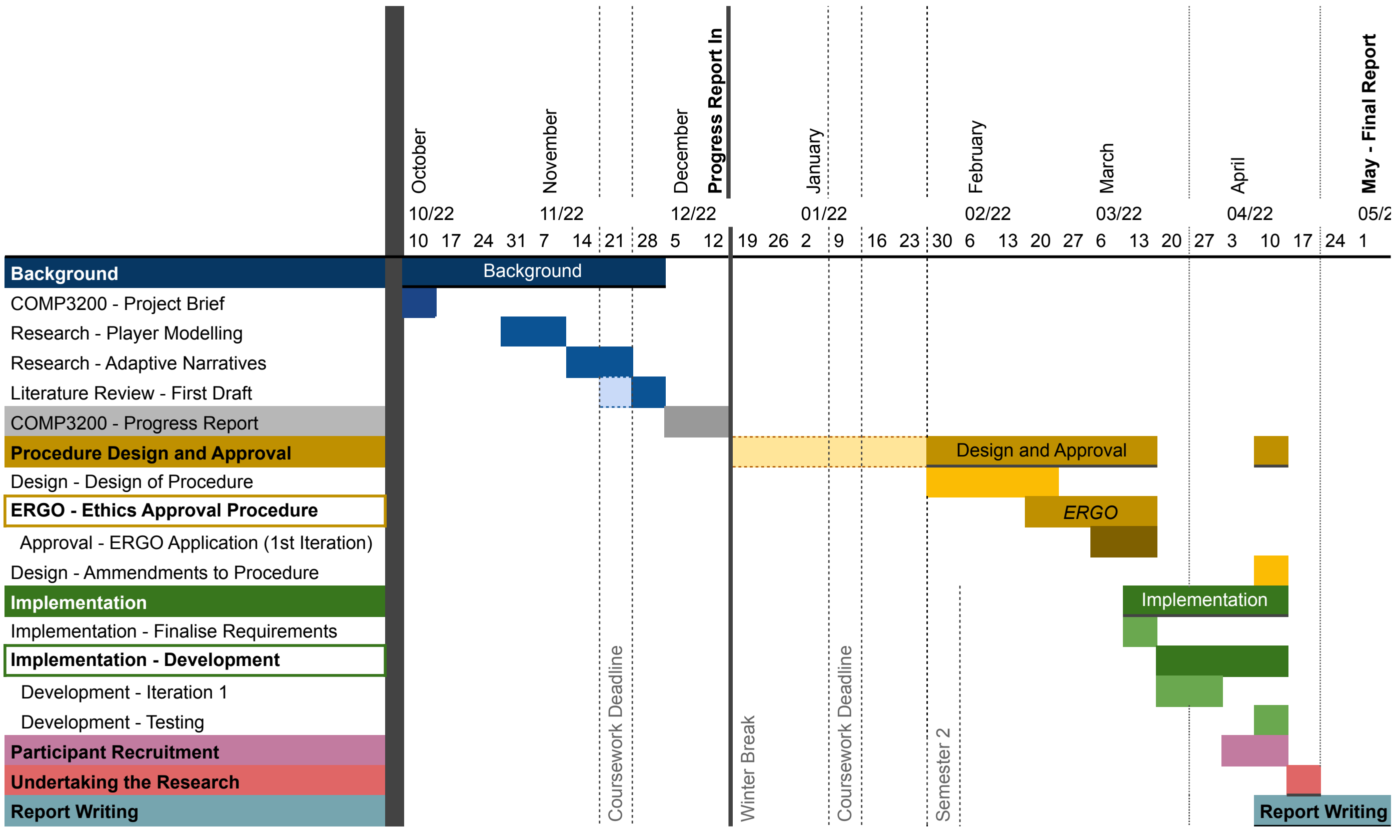

Figure 6.2: Final Gantt chart.

The final Gantt chart greatly differs from the original one. Due to a set of challenges (outlined in section 6.4), there was a delay in submitting the ethics application for this project. To accommodate this, only one iteration of development was held. Fortunately, a second iteration of ERGO was not needed, reducing the impact of the delay.

6.4 Addressing Challenges

Challenge 1: Delays
Care has been put into managing this project and despite passion behind the subject matter, I have to acknowledge a period of time in which I found it very difficult to focus and progress with the design for this project between deadlines for other modules and personal circumstances. This caused a severe delay which is reflected in the timeline (Figure 6.2). This meant that the experimental portion of the study had to be held throughout the Easter holidays, during which many students might not be on campus for what was originally an in-person study. To address this, the study was adapted so it could be held remotely. This way, participants could contribute regardless of their location by logging into a virtual machine through Remote Desktop.

Challenge 2: GPUs
When trying to test my game on the University's VM service, the machines did not have GPUs and hence were not able to run the game. In addition, I discovered that Remote Desktop was not designed for smooth playback. After experimenting with numerous remote-work software such as Parsec, it was decided that participants would need to download and run the game directly on their machines (requiring an ERGO amendment). This solution was not perfect and added a constraint for participant selection, but enabled the possibility of this study.

Challenge 3: Data Analysis Software
IBM's SPSS was originally planned to be utilised for quantitative data analysis. After this was proven to not be possible due to constraints with the trial version of the software, the study instead used an online tool provided on the London Psychometric Lab website. The tool does not provide evidence of provided calculations hence was not the first choice, but result-wise is functionally identical to the SPSS script .

7.1 Administrative Procedure

Potential participants were invited to express interest via email, at which point an informational document and consent form were sent to them. Upon return of signed consent and once an agreed date and time for the trial were decided, participation details were finalised within a password-protected spreadsheet (Figure 7.1). The group and pseudonym of each contestant alternate depending on their order of participation. At no point was a participant's group allocated based on other factors to reduce subconscious biases from the researcher.

Participant	Email	Consent Form Signed?	Agreed booked study time	Group
Participant "E1"	[illegible]	Yes	[illegible]	A
Participant "C1"	[illegible]	Yes	[illegible]	B
Participant "E2"	[illegible]	Yes	[illegible]	A
Participant "C2"	[illegible]	Yes	[illegible]	B
Participant "E3"	[illegible]	Yes	[illegible]	A
Participant "C3"	[illegible]	Yes	[illegible]	B
Participant "E4"	[illegible]	Yes	[illegible]	A
Participant "C4"	[illegible]	Yes	[illegible]	B
Participant "E5"	[illegible]	Yes	[illegible]	A
Participant "C5"	[illegible]	Yes	[illegible]	B

Figure 7.1: Spreadsheet holding participant details.

For any interviews, rough notes were taken and formatted for readability. Such notes can be found within the project archive:

Appendix A	Archive Table of Contents, including formatted interview notes.

7.2 Quantitative Analysis

7.2.1 Data Processing

Following the procedure set out in subsection 3.3.3, all survey data was processed and 5 factor values (well-being, self-control, emotionality, sociability and global trait EI) were generated from each survey. Each participant completed two surveys, and the differences between the two were calculated:

Appendix F 📂	Original and Processed Survey Scores

7.2.2 Hypothesis Testing

A two-sample T-test with the following parameters was performed on the processed data. A summary of the results and whether each difference was deemed significant are outlined below for each factor:

Factor	**p-value**	**Outcome at Significance Level = 0.10**
Global Trait EI	0.07603	**SIGNIFICANTLY DIFFERENT** **(Factor DECREASE in the experimental group)**
Well-being	0.29145	**NOT SIGNIFICANTLY DIFFERENT**
Self-Control	0.09598	**SIGNIFICANTLY DIFFERENT** **(Factor DECREASE in the experimental group)**
Emotionality	0.52057	**NOT SIGNIFICANTLY DIFFERENT**
Sociability	0.00839	**SIGNIFICANTLY DIFFERENT** **(Factor INCREASE in the experimental group)**

Details of the above hypothesis tests are provided in the following appendix:

Appendix G 📂	Detailed Hypothesis Tests

7.3 Qualitative Analysis

Ten semi-structured interviews were conducted throughout the duration of the study and notes were produced. Those notes were then reviewed to find any overarching themes and axially coded in accordance with any shared threads:

Themes present in interview

Participant	T1	T2	T3	T4	T5	T6	T7	T8	T8a	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	T19	T20	T21
E1	■	■	■	■		■	■	■				■					■		■	■		
E2					■			■	■			■		■	■							
E3								■	■						■	■	■	■				
E4	■						■	■							■	■		■	■	■		
E5													■			■	■				■	
C1								■														■
C2		■				■		■	■						■		■			■		
C3			■				■			■					■			■		■		
C4			■				■	■	■									■		■	■	
C5											■				■	■						
Total	2	2	3	1	1	2	4	**7**	4	1	1	2	1	1	**6**	4	4	4	2	**5**	2	1

T1	Questions were not representative of what the participant would actually do in reality	2
T2	Participant experienced an element of fear during the study	2
T3	Participant experienced an element of empathy during the study	3
T4	Participant liked the in-game characters	1
T5	Participant disliked the in-game characters	1
T6	Participant talked about the in-game mechanics	2
T7	Participant misunderstood the point of a room or series or rooms - showed general confusion	4
T8	Participant did not think their initial answers impacted the game	**7**
T8a	Participant forgot the questions or their answers to the initial questionnaire	4
T9	Participant thought their initial answers partially impacted the game	1
T10	Participant strongly believed their initial answers impacted the game	1
T11	Participant stated that they did not explore in the first room	2
T12	Participant stated they they explored the first room	1
T13	Participant mentioned their sadness when excluded by the NPCs	1
T14	Participant correctly identified the intent of a room	**6**
T15	Participant expressed that they had fun playing the game	4
T16	Participant experimented with the mechanics	4
T17	Participant showed interest in the details of the procedure and theory behind the study	4
T18	Participant noticed the music change in the Ball Throwing Room	2
T19	Participant noticed/commented on the facial reactions of the NPCs	**5**
T20	Participant did not believe that the game had any effect on them	2
T21	Participant mentioned that there should have been more questions at the introduction	1

Figure 7.2: Tables showing emergent themes and coded interviews.

8.1 Study Evaluation

Four goals were outlined for this study in chapter 4:

Study Goal	Status	Evidence
G1: Summarise existing literature.	Achieved	chapter 2
G2: Design and run a study to gather quantitative and qualitative data which can be used to accept/reject H_1 .	Achieved	chapter 3
G3: Process, present and evaluate the data using appropriate methods.	Achieved	chapter 7
G4: Use the data to accept/reject H_1 and draw a conclusion.	Achieved	section 8.3

The four goals were achieved with full completion, and this report concludes the study to be successful.

8.2 Discussion

This discussion will provide a grounded examination of the results gathered throughout this study, highlight prominent observations and offer commentary on suggested drawbacks of this study.

During the research portion of the project, several connections were found spanning from player modelling to emotional intelligence (see subsection 3.1.2), and a video game was designed in an attempt to explore these connections and examine the effect that personalisation and player modelling have on the ability of a game to evoke change in emotional intelligence. Admittedly, the hoped outcome was that this study would show that personalisation and player modelling have a positive effect on this quality of video games, however the nature of the effect was kept intentionally open to either a positive **or negative** effect.

In fact, some instances of gathered data point exactly to that: hypothesis tests concluded that participants in the experimental group actually showed a significant decrease in Global Trait EI if taken as a whole and in self-control if analysing the factors separately. Though the study acknowledges the relatively low sample size ($n_1 = 5, n_2 = 5$), and encourages the results of the hypothesis tests to be viewed through a lens of holistic scepticism, a possible decrease in trait EI could have arisen due to the design of the game as opposed to the underlying theory. Throughout the interviews, a general theme of confusion was brought up: four out of ten participants commented that at some point during their play they were not sure what to do, and one participant was not sure how the surveys at the end were linked to the game.

Overall, seven out of the ten participants did not believe that the introductory quiz had an effect on their game irrespective of their group. I believe this is a drawback of procedure design: participants were expecting more granular changes within each room and were not briefed that the programmed change was actually the selection of which rooms were presented.

Despite the uncertainty on the effect of personalisation on emotional intelligence development, qualitative analysis highlighted two important themes. Firstly, despite the actual effectiveness of the rooms, six out of ten participants expressed an understanding of what the rooms were trying to achieve - this provides hope for future iterations of this project. Secondly, a significant number of participants mentioned that they paid close attention to the facial reactions of the NPCs - this phenomenon can act as a basis for future room designs or studies which discuss empathy between human and non-playable characters.

8.3 Conclusion

Neither the survey nor the interview data show that personalisation and player modelling have an indisputable effect on the capacity of video games for emotional development, however it is strongly suggested that comes as a product of study design and should not be used to disprove the theoretical connections outlined in this study's concept map. Despite this, it is believed that findings from the study may provide valuable insights into future research done on affective video games.

Future work stemming from this study should seek to iterate on the provided video game design to better translate the theory outlined in the concept map into a more effective research tool, particularly in communicating the expectations of the player. In order to focus on the personalisation aspect of the study, a future tool should redesign what it presents to the control group. One suggestion is to design a set of 'neutral' rooms for these players, instead of providing purposeful rooms at random. When designing affective rooms, a future tool should see if it can focus on facial expressions of NPCs as opposed to their actions, as this was seen as more effective.

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[53] J. Sibthorp, K. Paisley, J. Gookin, and P. Ward, "Addressing Response-shift Bias: Retrospective Pretests in Recreation Research and Evaluation," *Journal of Leisure Research*, vol. 39, no. 2, pp. 295–315, Jun. 2007, ISSN: 0022-2216. DOI: 10.1080/00222216.2007.11950109. [Online]. Available: https://doi.org/10.1080/00222216.2007.11950109 (visited on 03/02/2023).

The project archive contains supporting materials and source files from the duration of the project:

```
archive.zip
└─ comp3200-rp1g20
   ├─ ethics-and-data-protection
   │  ├─ consent-form.pdf
   │  ├─ feps-dpa-plan.pdf
   │  └─ participant-info-sheet.pdf
   ├─ interview-notes
   │  └─ interview-notes.pdf
   ├─ latex-source
   │  ├─ appendices
   │  │  └─ ...
   │  ├─ figures
   │  │  └─ ...
   │  ├─ references.bib
   │  └─ report.tex
   ├─ surveys
   │  ├─ original-teique-sf.pdf
   │  ├─ post-survey.pdf
   │  └─ retrospective-pre-survey.pdf
   └─ unity-project-source
      └─ ...
```

A copy of archive.zip is available from The ECS Part III Project Archive, or by request from R.Popovs@soton.ac.uk

Appendix B Verifiable Word Count

Word count generated with
`texcount -merge -sub=chapter -sum=1,1,1 report.tex`

Contents of generated word count:

```
Subcounts
Chapter: Project Introduction: 457
Chapter: Literature Review: 2795
Chapter: Methodology and Design: 3466
Chapter: Goals and Requirements: 423
Chapter: Game Development and Testing: 1225
Chapter: Project Management: 592
Chapter: Conducting the Study and Data Analysis: 325
Chapter: Evaluation and Conclusion: 703


Total Word count (text+headers+other)
9986 (9277+168+541)
```

Question 1: You are working on a piece of group coursework however some members of the group are not being proactive, causing overdue tasks.		
Do you...	$\Delta_{\text{Soc.}}$	$\Delta_{\text{Ach.}}$
Try and get the members involved by booking a group meeting	+1	0
Leave it and hope that the other group members sort it out between themselves	-1	0
Take up the tasks and do them yourself to save time	-1	+1

Question 2: Congratulations! You have applied for a job you are very passionate about and you start next week! The job policy allows you to go home early if you have completed all of your tasks.		
Do you...	$\Delta_{\text{Soc.}}$	$\Delta_{\text{Ach.}}$
Do your tasks to an expected level and go home a few hours early	-1	-1
Try and perfect the solutions to your tasks even if it takes slightly longer	+1	+1
Finish your tasks as soon as possible so you can start something new	0	-1

Question 3: You have been invited to a 'haunted house' experience by your friends. This experience poses no real danger but involves actors dressed up in scary costumes. Before going into the experience, one of your friends admits that they are anxious regarding the actors.		
Do you...	$\Delta_{\text{Soc.}}$	$\Delta_{\text{Ach.}}$
Play along with the actors during the experience	-1	0
Stay by your friend throughout the experience	+1	0
Remind your friend that the characters are not real throughout the experience	+1	0

Question 4: You are placed within a boring-looking room and are told that there are multiple secrets within it. Finding those secrets does not award you anything and you have the option of skipping the room entirely and moving on.
Do you...
Skip the room
Search the room for secrets
At this point, if the player actively walks around the room they are awarded +1 $\Delta_{\text{Exp.}}$ which overwrites the effects of either choice that they then pick Each question can be answered by interacting with a physical button within the game world or via quick keyboard shortcuts. This will be mentioned to the player. If the player does not choose to use the physical buttons for any of the questions, they will be deducted -1 $\Delta_{\text{Imm.}}$

Appendix D Permission to adapt the TEIQue-SF

file:///C:/Users/onyxi/Desktop/ERGOII%20-%20Attempt%

From: Ross Popovs (rp1g20) <rp1g20@soton.ac.uk>
Sent: 02 March 2023 18:05
To: admin@teique.com
Subject: TEIQue for undergraduate thesis: digitisation and question altering

Greetings,

My name is Ross and I am undergraduate student at the University of Southampton. I am looking to use TEIQue-SF in my undergraduate thesis twice, as a retrospective pre-test and as a post-test.

I was wondering whether it is within policy for me to alter the questions in the first test to make them past tense (Example from “I tend to change my mind frequently” to “In the past, I have tended to change my mind frequently”). I do not intend to change the nature of the questions in any other way.

I was also wondering if I am allowed to ask these questions via an online form instead of a physical handout.

Credit and referencing will of course be provided.
Thank you offering this resource for academic use!

Best regards,
Ross Popovs
University of Southampton

From: admin@teique.com
Sent: 03 March 2023 09:04
To: Ross Popovs (rp1g20)
Subject: RE: TEIQue for undergraduate thesis: digitisation and question altering

Dear Mr Popovs

Thank you kindly for your email. You do not need special permission to use any TEIQue form in academic research, provided it is strictly for non-commercial purposes and no individualized feedback is given to respondents. Please also see our FAQ at http://psychometriclab.com/faq/ In particular, note that we do not provide free access to norms or feedback reports. We encourage a donation of £69.99 for the TEIQue and £49.99 for the TEIQue-SF. The donation button is labelled "Support" on the landing page at www.psychometriclab.com.

You can download the various TEIQue forms (including translations and adaptations) from the same website, which also incorporates an automated on-line scoring system for the TEIQue and TEIQue-SF. The scoring key for the full form of the TEIQue is not publicly available due to various licensing agreements in place. The scoring key for the TEIQue-SF and TEIQue-ASF is exactly the same and both forms can be scored via the online scoring engine that is available on the website (www.psychometriclab.com). You can also download this scoring key from http://psychometriclab.com/scoring-the-teique/

Regarding online usage, it is OK, provided that you remove the instrument immediately after the end of the project and include the following copyright notice and weblink in a salient position on the data collection form.

Copyright © K. V. Petrides 1998 – 2023
www.psychometriclab.com

Please note that the forms are to be used exclusively for academic or medical research purposes. Further note that we cannot provide additional support beyond what is already on our websites although you can join us in one of our Social Media below.

We hope this helps and wish you good luck with your research,

The TEIQue Team

----

https://www.psychometriclab.com

https://www.linkedin.com/in/kvpetrides

https://www.ucl.ac.uk/pals/people/kv-petrides

https://www.teique.com

https://www.traitEI.com

https://www.facebook.com/traitemotionalintelligence

https://www.instagram.com/teique_com

SPSS Script provided by the London Psychometric Lab
(Sourced from https://psychometriclab.com/obtaining-the-teique/) [50]

```
RECODE
  tei_16 tei_2 tei_18 tei_4 tei_5 tei_7 tei_22 tei_8 tei_10 tei_25
  tei_26 tei_12 tei_13 tei_28 tei_14  (7=1)  (6=2)  (5=3)
   (3=5)  (2=6)  (1=7)  .
EXECUTE .

COMPUTE tot_tei = (tei_1 +tei_2+tei_3+tei_4+tei_5+tei_6+tei_7
+tei_8+tei_9+tei_10+tei_11+tei_12+tei_13+tei_14+tei_15+tei_16
+tei_17+tei_18+tei_19+tei_20+tei_21+tei_22+tei_23+tei_24+tei_25
+tei_26+tei_27+tei_28+tei_29+tei_30)/30 .
EXECUTE .

*Factor scores .
COMPUTE well_being = (tei_5+ tei_20+ tei_9 +tei_24+
  tei_12 +tei_27)/6.
EXECUTE .
COMPUTE self_control = (tei_4+ tei_19+ tei_7 +tei_22
  +tei_15+ tei_30)/6 .
EXECUTE .
COMPUTE emotionality = (tei_1+ tei_16+ tei_2 +tei_17+
  tei_8+ tei_23+ tei_13+ tei_28)/8  .
EXECUTE .
COMPUTE sociability = (tei_6 +tei_21+ tei_10+ tei_25
  +tei_11+ tei_26)/6 .
EXECUTE .

var lab tot_tei 'global trait emotional intelligence'  .

TITLE 'well_being' .
RELIABILITY
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  /SCALE(ALPHA)=ALL/MODEL=ALPHA
  /SUMMARY=TOTAL .

TITLE 'self-control' .
RELIABILITY
  /VARIABLES= tei_4 tei_19 tei_7 tei_22 tei_15 tei_30
  /FORMAT=NOLABELS
  /SCALE(ALPHA)=ALL/MODEL=ALPHA
  /SUMMARY=TOTAL .

TITLE 'emotionality' .
RELIABILITY
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  /FORMAT=NOLABELS
  /SCALE(ALPHA)=ALL/MODEL=ALPHA
  /SUMMARY=TOTAL .

TITLE 'sociability' .
RELIABILITY
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  /FORMAT=NOLABELS
```

```
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  /SUMMARY=TOTAL .

TITLE 'global trait EI' .
RELIABILITY
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    tei_9 tei_10 tei_11 tei_12 tei_13 tei_14 tei_15 tei_16 tei_17
    tei_18 tei_19 tei_20 tei_21 tei_22 tei_23 tei_24 tei_25
    tei_26 tei_27 tei_28 tei_29 tei_30 .
  /FORMAT=NOLABELS
  /SCALE(ALPHA)=ALL/MODEL=ALPHA
```

Appendix F Original and Processed Survey Scores

Control Group - Generated Factors:

Retrospective Pre-Survey

	Global score	Well-being	Self-Control	Emotionality	Sociability
C1	3.266667	3.5	3	3.625	2.666667
C2	5.033333	5.666667	4.666667	6.5	3.833333
C3	4.366667	5	2.666667	4.375	5
C4	4.8	4	3.333333	5.75	5.833333
C5	4.266666667	3.833333333	4	4.375	4.5

Post-Survey

	Global score	Well-being	Self-Control	Emotionality	Sociability
C1	3.4	3.833333	3.333333	3.5	2.333333
C2	5.3	6.333333	4.666667	6.5	3.5
C3	4.633333	5.833333	3	4.25	5.333333
C4	4.566667	3.333333	4	5.25	5.666667
C5	4.366666667	4.166666667	5.166666667	4.25	4.5

Experimental Group - Generated Factors:

Retrospective Pre-Survey

	Global score	Well-being	Self-Control	Emotionality	Sociability
E1	4.7	6.166667	3.5	4.625	4.333333
E2	3.8	4.166667	2	4	5
E3	4.166667	5.5	4.666667	3.375	4.666667
E4	4.866667	5.333333	5.166667	5.25	3.833333
E5	3.366667	4.833333	2.833333	2.875	2.333333

Post-Survey

	Global score	Well-being	Self-Control	Emotionality	Sociability
E1	4.5	6.166667	3.5	4.375	4
E2	3.9	4	2.333333	4.125	4.666667
E3	4.633333	5.5	5.166667	3.875	5
E4	4.833333	5.5	5.333333	4.75	3.833333
E5	3.3	4.833333	2.666667	2.625	2.833333

Control Group - Factor Differences:

	Trait EI Score Differences (Control Group)				
	Global score	Well-being	Self-Control	Emotionality	Sociability
C1	0.133333	0.333333	0.333333	-0.125	-0.33333
C2	0.266667	0.666667	0	0	-0.33333
C3	0.266667	0.833333	0.333333	-0.125	0.333333
C4	-0.23333	-0.66667	0.666667	-0.5	-0.16667
C5	0.1	0.333333337	1.166666667	1.625	0

Experimental Group - Factor Differences:

	Trait EI Score Differences (Experimental Group)				
	Global score	Well-being	Self-Control	Emotionality	Sociability
E1	-0.2	0	0	-0.25	-0.33333
E2	0.1	-0.16667	0.333333	0.125	-0.33333
E3	0.466667	0	0.5	0.5	0.333333
E4	-0.03333	0.166667	0.166667	-0.5	0
E5	-0.06667	0	-0.16667	-0.25	0.5

$$\bar{x_i} = \frac{\sum x}{n_i}$$

$$S_i = \sqrt{\frac{\sum(x - \bar{x_i})^2}{n_i}}$$

$$Sp = \sqrt{\frac{(n_1 - 1){S_1}^2 + (n_2 - 1){S_2}^2}{n_1 + n_2 - 2}}$$

$$t = \frac{\bar{x_1} - \bar{x_2}}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Two-Sample T-Test (Global Trait EI)	
Sample Size n_1 **= 5**	Sample Size n_2 **= 5**
$\bar{x}_1$ (Mean of CG differences) = 0.2666685	$\bar{x}_2$ (Mean of EG differences) = 0.0533334
S_1 (S.D of CG differences) = 0.2046665	S_2 (S.D of EG differencess) = 0.1138225
$\mathbf{H_0}$: $\mu_1 = \mu_2$ *i.e. population means are equal*	$\mathbf{H_1}$: $\mu_1 \neq \mu_2$ *i.e. population means are significantly different*
Significance Level α = 0.10	Degrees of freedom = $n_1 + n_2 - 2$ = 8
Calculating the test statistic: t = 2.03696	
Calculating the p-value of the statistic: p = **0.07603** **The result is SIGNIFICANT at p $<$ 0.10**	

Two-Sample T-Test (Well-being)	
Sample Size n_1 **= 5**	Sample Size n_2 **= 5**
$\bar{x}_1$ (Mean of CG differences) = 0.2999993	$\bar{x}_2$ (Mean of EG differences) = 0
S_1 (S.D of CG differences) = 0.5821430	S_2 (S.D of EG differencess) = 0.1178524
$\mathbf{H_0}$: $\mu_1 = \mu_2$ *i.e. population means are equal*	$\mathbf{H_1}$: $\mu_1 \neq \mu_2$ *i.e. population means are significantly different*
Significance Level α = 0.10	Degrees of freedom = $n_1 + n_2 - 2$ = 8
Calculating the test statistic: t = 1.12941	
Calculating the p-value of the statistic: p = **0.29145** **The result is NOT SIGNIFICANT at p $<$ 0.10**	

Two-Sample T-Test (Self-Control)	
Sample Size n_1 **= 5**	Sample Size n_2 **= 5**
$\bar{x}_1$ (Mean of CG differences) = 0.4999999	$\bar{x}_2$ (Mean of EG differences) = 0.06666
S_1 (S.D of CG differences) = 0.4409586	S_2 (S.D of EG differencess) = 0.2635241
$\mathbf{H_0} : \mu_1 = \mu_2$ *i.e. population means are equal*	$\mathbf{H_1} : \mu_1 \neq \mu_2$ *i.e. population means are significantly different*
Significance Level α = 0.10	Degrees of freedom = $n_1 + n_2 - 2$ = 8
Calculating the test statistic: t = 1.886265	
Calculating the p-value of the statistic: p = **0.09598** **The result is SIGNIFICANT at p $<$ 0.10**	

Two-Sample T-Test (Emotionality)	
Sample Size n_1 **= 5**	Sample Size n_2 **= 5**
$\bar{x}_1$ (Mean of CG differences) = 0.175000	$\bar{x}_2$ (Mean of EG differences) = -0.0750000.
S_1 (S.D of CG differences) = 0.831978	S_2 (S.D of EG differencess) = 0.391312
$\mathbf{H_0} : \mu_1 = \mu_2$ *i.e. population means are equal*	$\mathbf{H_1} : \mu_1 \neq \mu_2$ *i.e. population means are significantly different*
Significance Level α = 0.10	Degrees of freedom = $n_1 + n_2 - 2$ = 8
Calculating the test statistic: t = 0.671914	
Calculating the p-value of the statistic: p = **0.52057** **The result is NOT SIGNIFICANT at p $<$ 0.10**	

Two-Sample T-Test (Sociability)	
Sample Size n_1 **= 5**	Sample Size n_2 **= 5**
$\bar{x}_1$ (Mean of CG differences) = -0.099999	$\bar{x}_2$ (Mean of EG differences) = 0.0333346
S_1 (S.D of CG differences) = 0.278873	S_2 (S.D of EG differencess) = 0.380042
$\mathbf{H_0} : \mu_1 = \mu_2$ *i.e. population means are equal*	$\mathbf{H_1} : \mu_1 \neq \mu_2$ *i.e. population means are significantly different*
Significance Level α = 0.10	Degrees of freedom = $n_1 + n_2 - 2$ = 8
Calculating the test statistic: t = -3.47457	
Calculating the p-value of the statistic: p = **0.00839** **The result is SIGNIFICANT at p $<$ 0.10**	